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Stapf et al.

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(54) **TWO COMPONENT (2K) CURABLE
ADHESIVE COMPOSITION**

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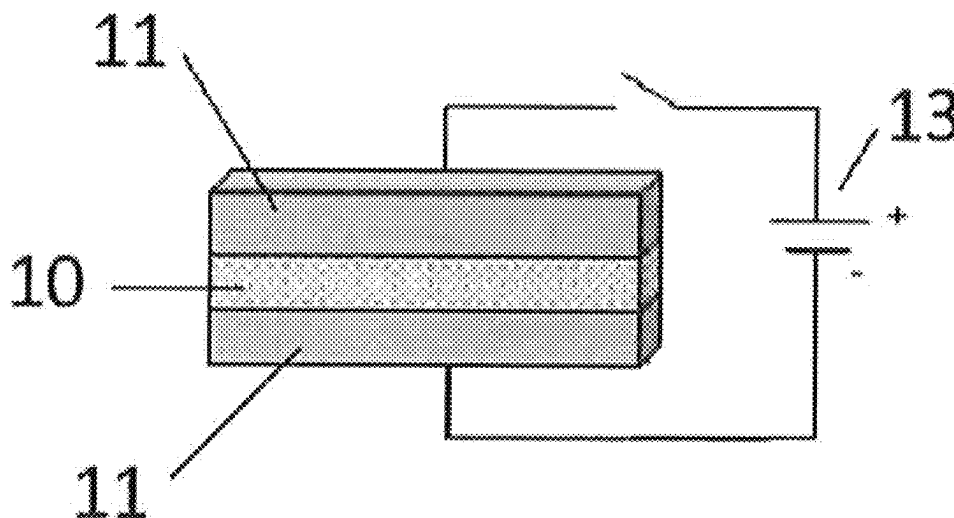
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(57) **ABSTRACT**

The present invention is directed to a curable and debond-
able two-part (2K) adhesive composition comprising: i) a
first part comprising: a) epoxy resin; b) an electrolyte; c)
optionally, a solubilizer; and, ii) a second part comprising: a)
a curing agent consisting of at least one compound possess-
ing at least two epoxide reactive groups per molecule; and,
b) an accelerator; wherein said composition further com-
prises an electrically non-conductive filler and, optionally a
toughener.

20 Claims, 4 Drawing Sheets



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See application file for complete search history.

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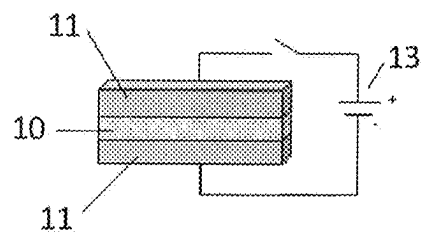


Fig. 1a

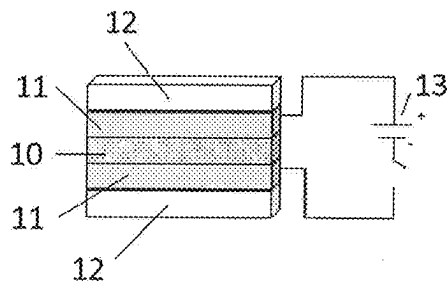


Fig. 1b

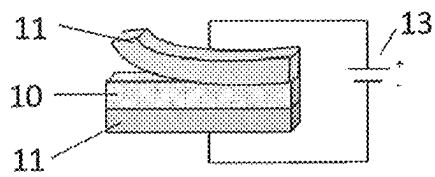


Fig. 2a

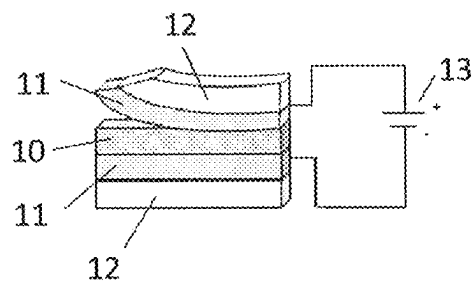


Fig. 2b

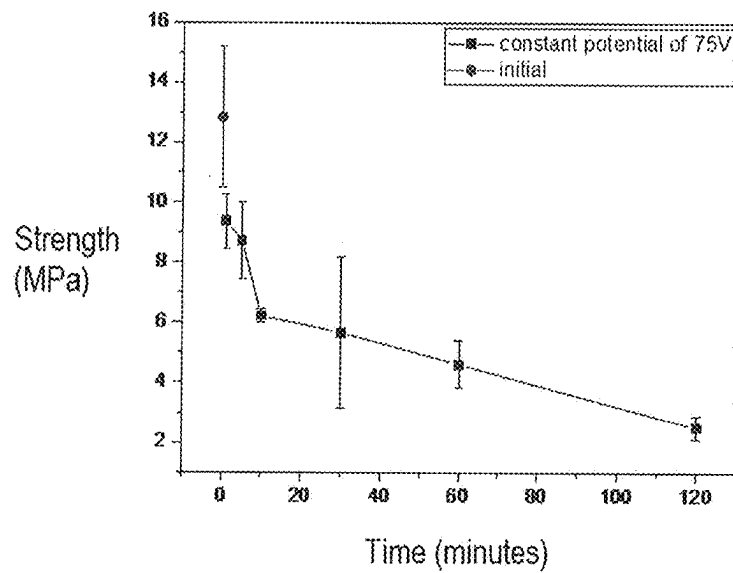


Fig. 3a

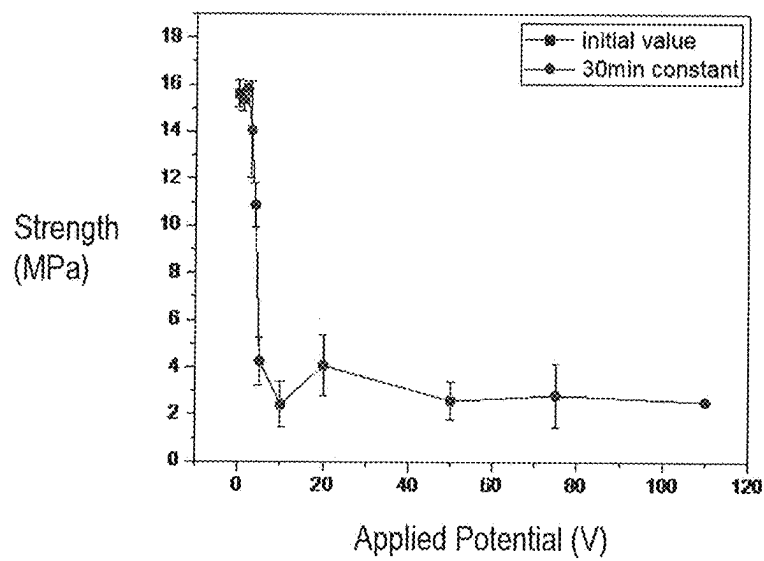


Fig. 3b

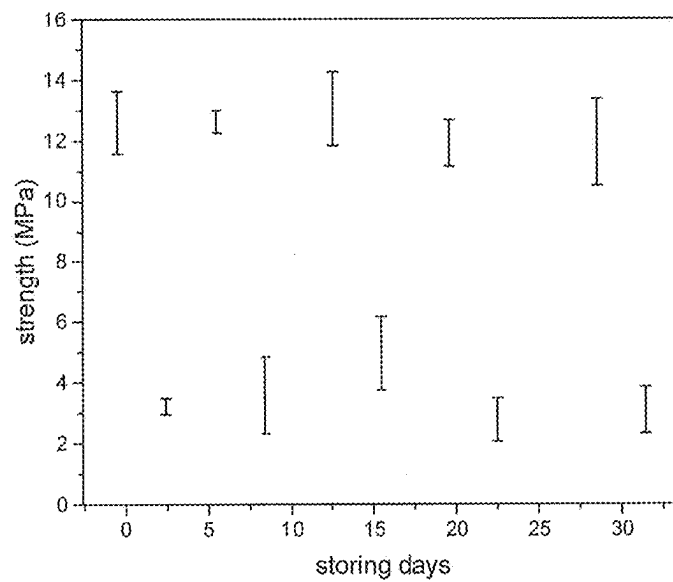


Fig. 4

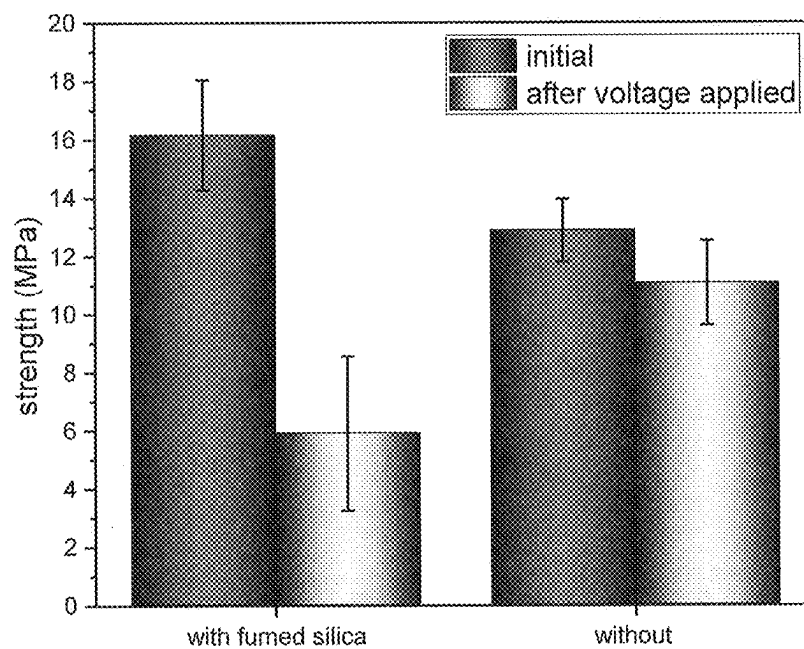


Fig. 5

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TWO COMPONENT (2K) CURABLE ADHESIVE COMPOSITION

FIELD OF THE INVENTION

The present invention is directed to an adhesive composition which can be debonded from particular substrates to which it is applied. More particularly, the present invention is directed to a two-part (2K) curable and debondable adhesive composition.

BACKGROUND TO THE INVENTION

Adhesive bonds and polymeric coatings are commonly used in the assembly and finishing of manufactured goods. They are used in place of mechanical fasteners, such as screws, bolts and rivets, to provide bonds with reduced machining costs and greater adaptability in the manufacturing process. Adhesive bonds distribute stresses evenly, reduce the possibility of fatigue and seal the joints from corrosive species.

Whilst adhesive bonds thus offer many advantages over mechanical fasteners, it tends to be difficult to disassemble adhesively bonded objects where this is required in practical applications. The removal of the adhesive through mechanical processes—such as by sand blasting or by wire brushing—is often precluded, in part because the adhesive is disposed between substrates and is thus either inaccessible or difficult to abrade without corrupting the substrate surfaces. Disassembly through the application of chemicals and/or high temperature—such as disclosed in U.S. Pat. No. 4,171,240 (Wong) and U.S. Pat. No. 4,729,797 (Linde et al.)—might be effective but can be time consuming and complex to perform; moreover, the aggressive chemicals and/or harsh conditions required can damage the substrates being separated, rendering them unsuitable for subsequent applications.

Noting these problems, certain authors have sought to develop debondable adhesive compositions, wherein the passage of an electrical current through the cured compositions acts to disrupt the bonding at the interface of the adhesive and the substrate.

U.S. Pat. No. 7,465,492 (Gilbert) describes an disbondable composition comprising: a matrix functionality comprising a monomer selected from the group consisting of acrylics, methacrylics and combinations thereof; a free radical initiator; and, an electrolyte, wherein the electrolyte provides sufficient ionic conductivity to said composition to support a faradaic reaction at a bond formed between the composition and an electrically conductive surface and thus allows the composition to disbond from the surface.

US 2007/0269659 (Gilbert) describes an adhesive composition disbondable at two interfaces, the composition: (i) comprising a polymer and an electrolyte; (ii) facilitating joinder of two surfaces; and, (iii) in response to a voltage applied across both surfaces so as to form an anodic interface and a cathodic interface, disbonding from both the anodic and cathodic surfaces.

US 2008/0196828 (Gilbert) describes a hot-melt adhesive composition comprising: a thermoplastic component; and, an electrolyte, wherein the electrolyte provides sufficient ionic conductivity to the composition to enable a faradaic reaction at a bond formed between the composition and an electrically conductive surface and to allow the composition to disbond from the surface.

WO2017/133864 (Henkel AG & Co. KGaA) describes a method for reversibly bonding a first and a second substrate,

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wherein at least the first substrate is an electrically non-conductive substrate, the method comprising: a) coating the surface of the electrically non-conductive substrate(s) with a conductive ink; b) applying an electrically debondable hot melt adhesive composition to the conductive ink-coated surface of the first substrate and/or the second substrate; c) contacting the first and the second substrates such that the electrically debondable hot melt adhesive composition is interposed between the two substrates; d) allowing formation of an adhesive bond between the two substrates to provide bonded substrates; and, e) applying a voltage to the bonded substrates whereby adhesion at an interface between the electrically debondable hot melt adhesive composition and a substrate surface is substantially weakened.

There remains a need in the art to provide an adhesive composition which can be conveniently applied to the surfaces of substrates to be bonded, which upon curing thereof can provide an effective bond within composite structures containing said substrates but which can be effectively de-bonded from those substrates by the facile application of an electrical potential across the cured adhesive.

STATEMENT OF THE INVENTION

In accordance with a first aspect of the invention, there is provided a curable and debondable two-part (2K) adhesive composition comprising:

a first part comprising:

- a) epoxy resin;
- b) an electrolyte;
- c) optionally, a solubilizer; and

a second part comprising:

- a) a curing agent consisting of at least one compound possessing at least two epoxide reactive groups per molecule; and,
- b) an accelerator; and,

wherein said composition further comprises an electrically non-conductive filler and, optionally a toughener.

In important embodiments of the invention, the two part (2K) adhesive composition comprises, based on the weight of the composition,

from 30 to 70 wt. %, preferably 35 to 60 wt. % of said epoxy resin;

from 2 to 25 wt. %, preferably 5 to 20 wt. % of said electrolyte;

from 0 to 15 wt. %, preferably from 1.5 to 10 wt. %, of solubilizer;

from 0.1 to 15 wt. %, preferably from 1 to 15%, of accelerator;

from 1 to 50 wt. %, preferably from 2 to 25%, of electrically non-conductive filler; and,

from 0 to 10 wt. %, preferably 0.1 to 5 wt. %, of electrically conductive filler.

In the first part of the adhesive composition, said electrolyte is preferably selected from the group consisting of 1-ethyl-3-methylimidazolium methansulfonate, 1-ethyl-3-methylimidazolium methyl sulfate, 1-hexyl-3-methylimidazolium 2-(2-fluoroanilino)-pyridinate, 1-hexyl-3-methylimidazolium imide, 1-butyl-1-methyl-pyrrolidinium 2-(2-fluoroanilino)-pyridinate, 1-butyl-1-methyl-pyrrolidinium imide, trihexyl (tetradecyl) phospholium 2-(2-fluoroanilino)-pyridinate, cyclohexyltrimethylammonium bis(trifluoromethylsulfonyl) imide, di(2-hydroxyethyl) ammonium trifluoroacetate, N,N-dimethyl (2-hydroxyethyl) ammonium octanoate, methyltriocylammonium bis(trifluoromethylsulfonyl) imide, N-ethyl-N—N—N—N-tetramethylguanidinium trifluoromethanesulfonate, guanidinium trifluoro-

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romethanesulfonate, 1-butyl-4-methylpyridinium bromide, 1-butyl-3-methylpyridinium tetrafluoroborate, 1-butyl-3-hydroxymethylpyridinium ethylsulfate, 1-butyl-1-methylpyrrolidinium bis(trifluoromethylsulfonyl) imide, 1-butyl-methylpyrrolidinium tris(pentafluoroethyl) 5 trifluorophosphate, 3-methyl imidazolium ethylsulfate, 1-ethyl-3-methylimidazolium chloride, 1-ethyl-3-ethyl-methylimidazolium bromide, 1-butyl-3-methylimidazolium chloride, 1-hexyl-3-methylimidazolium chloride, 1-octyl-3-methylimidazolium chloride, 1-methyl-3-octylimidazolium chloride, 1-propyl-3-methylimidazolium iodide, 1-butyl-3-methylimidazolium tetrafluoroborate, 1-butyl-3-methylimidazolium trifluoromethanesulfonate, 1-butyl-3-methylimidazolium hexafluorophosphate, 1-butyl-2,3-dimethylimidazolium tetrafluoroborate, 1-butyl-2,3-dimethylimidazolium hexafluorophosphate, 1-butylimidazol, 1-methylimidazolium tetrafluoroborate, tetrabutylphosphonium tris(pentafluoroethyl) trifluorophosphate, trihexyl(tetradecyl) phosphonium tetrafluoroborate and mixtures thereof. A particular preference for the use of at least one of 1-ethyl-3-methylimidazolium methanesulfonate, 1-ethyl-3-methylimidazolium methyl sulfate may be mentioned.

Independently of or additional to this statement of preference for the electrolyte in the first part, it is preferred that said composition comprises a solubilizer selected from: polyphosphazenes; polymethylenesulfides; polyoxyalkylene glycols; polyethylene imines; silicone surfactants and fluorinated silicone surfactants; co-polymers of functionalized polyalkylsiloxanes and epoxy resins; polyhydric alcohols; and, sugars.

In the second part of the composition, it is preferred that said curing agent comprises at least one polyamine having at least two amine hydrogens reactive toward epoxide groups, said polyamine being further characterized by containing primary and/or secondary amine groups and having an equivalent weight per primary or secondary amine group of not more than 150 g/eq. A particular preference for the use of ether group-containing aliphatic primary polyamines may be mentioned in this regard.

Independently of or additional to this statement of preference for the curing agent of the second part of the composition, it is preferred that said accelerator is selected from the group consisting of tertiary amines, quaternary ammonium salts, amidines, guanidines and mixtures thereof. A particular preference is noted for employing at least one accelerator selected from imidazole, methylimidazole, benzyl dimethylamine, 1,8-diazabicyclo[5.4.0]undec-7-ene and 1,4-diazabicyclo(2,2,2) octane.

In accordance with a second aspect of the invention, there is provided a bonded structure comprising:

- a first material layer having an electrically conductive surface; and,
- a second material layer having an electrically conductive surface,

wherein the cured and debondable two-part (2K) adhesive composition as defined hereinabove and in the appended claims is disposed between said first and second material layers.

In accordance with a third aspect of the present invention, there is provided a method of debonding said bonded structure as defined hereinabove and in the appended claims, the method comprising the steps of:

- i) applying a voltage across both surfaces to form an anodic interface and a cathodic interface; and,
- ii) debonding the surfaces.

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Step i) of this method is preferably characterized by at least one of:

- a) an applied voltage of from 0.5 to 200 V; and,
- b) the voltage being applied for a duration of from 1 second to 120 minutes, preferably from 1 seconds to 60 minutes.

Definitions

As used herein, the singular forms “a”, “an” and “the” include plural referents unless the context clearly dictates otherwise.

The terms “comprising”, “comprises” and “comprised of” as used herein are synonymous with “including”, “includes”, “containing” or “contains”, and are inclusive or open-ended and do not exclude additional, non-recited members, elements or method steps.

As used herein, the term “consisting of” excludes any element, ingredient, member or method step not specified.

When amounts, concentrations, dimensions and other parameters are expressed in the form of a range, a preferable range, an upper limit value, a lower limit value or preferable upper and limit values, it should be understood that any ranges obtainable by combining any upper limit or preferable value with any lower limit or preferable value are also specifically disclosed, irrespective of whether the obtained ranges are clearly mentioned in the context.

Further, in accordance with standard understanding, a weight range represented as being “from 0 to x” specifically includes 0 wt. %: the ingredient defined by said range may be absent from the composition or may be present in the composition in an amount up to x wt. %.

The words “preferred”, “preferably”, “desirably” and “particularly” are used frequently herein to refer to embodiments of the disclosure that may afford particular benefits, under certain circumstances. However, the recitation of one or more preferable, preferred, desirable or particular embodiments does not imply that other embodiments are not useful and is not intended to exclude those other embodiments from the scope of the disclosure.

As used throughout this application, the word “may” is used in a permissive sense—that is meaning to have the potential to—rather than in the mandatory sense.

As used herein, room temperature is 23° C. plus or minus 2° C. As used herein, “ambient conditions” means the temperature and pressure of the surroundings in which the composition is located or in which a coating layer or the substrate of said coating layer is located.

“Two-part (2K) compositions” in the context of the present invention are understood to be compositions in which a first part (A) and a second part (B) must be stored in separate vessels because of their (high) reactivity. The two parts are mixed only shortly before application and then react, typically without additional activation, with bond formation and thereby formation of a polymeric network. Herein higher temperatures may be applied in order to accelerate the cross-linking reaction.

As used herein the term “debondable” means that, after curing of the adhesive, the bond strength can be weakened by at least 50% upon application of an electrical potential of from 10V-75V for a duration of from 1 s to 60 minutes. The cured adhesive is applied between two substrates which are bonded by said adhesive so that an electric current is running through the adhesive bond line. Bond strength is measured by Tensile Lap Shear (TLS) test performed at room temperature and based upon EN 1465:2009 (German version) *Based on Adhesives—Determination of tensile lap-shear*

strength of bonded assemblies. The bond overlapping area was 25 mm×10 mm with a bond thickness of about 150 μm.

As used herein, the term “monomer” refers to a substance that can undergo a polymerization reaction to contribute constitutional units to the chemical structure of a polymer. The term “monofunctional”, as used herein, refers to the possession of one polymerizable moiety. The term “polyfunctional”, as used herein, refers to the possession of more than one polymerizable moiety.

As used herein, the term “equivalent (eq.)” relates, as is usual in chemical notation, to the relative number of reactive groups present in the reaction.

The term “electrolyte” is used herein in accordance with its standard meaning in the art as a substance containing free ions which can conduct electricity by displacement of charged carrier species. The term is intended to encompass molten electrolytes, liquid electrolytes, semi-solid electrolytes and solid electrolytes wherein at least one of the cationic or anionic components of their electrolyte structure is essentially free for displacement, thus acting as charge carrier.

The curable adhesive compositions of the present invention and the cured adhesives obtained therefrom possess “electrolyte functionality” in that the adhesive material permits the conduction of ions, either anions, cations or both. The electrolyte functionality is understood to derive from the ability of the compositions and cured adhesives to solvate ions of at least one polarity.

As used herein, “(meth)acryl” is a shorthand term referring to “acryl” and/or “methacryl”. Thus the term “(meth)acrylamide” refers collectively to acrylamide and methacrylamide.

As used herein, “C₁-C_n alkyl” group refers to a monovalent group that contains 1 to n carbons atoms, that is a radical of an alkane and includes straight-chain and branched organic groups. As such, a “C₁-C₃₀ alkyl” group refers to a monovalent group that contains from 1 to 30 carbons atoms, that is a radical of an alkane and includes straight-chain and branched organic groups. Examples of alkyl groups include, but are not limited to: methyl; ethyl; propyl; isopropyl; n-butyl; isobutyl; sec-butyl; tert-butyl; n-pentyl; n-hexyl; n-heptyl; and, 2-ethylhexyl. In the present invention, such alkyl groups may be unsubstituted or may be substituted with one or more substituents such as halo, nitro, cyano, amido, amino, sulfonyl, sulfinyl, sulfanyl, sulfoxy, urea, thiourea, sulfamoyl, sulfamide and hydroxy. Where applicable, a preference for a given substituent will be noted in the specification. In general, however, a preference for alkyl groups containing from 1-18 carbon atoms (C₁-C₁₈ alkyl)—for example alkyl groups containing from 1 to 12 carbon atoms (C₁-C₁₂ alkyl) or from 1 to 6 carbon atoms (C₁-C₆ alkyl)—should be noted.

The term “C₁-C₁₈ hydroxyalkyl” as used herein refers to a HO-(alkyl) group having from 1 to 18 carbon atoms, where the point of attachment of the substituent is through the oxygen-atom and the alkyl group is as defined above.

An “alkoxy group” refers to a monovalent group represented by —OA where A is an alkyl group: non-limiting examples thereof are a methoxy group, an ethoxy group and an iso-propyloxy group. The term “C₁-C₁₈ alkoxyalkyl” as used herein refers to an alkyl group having an alkoxy substituent as defined above and wherein the moiety (alkyl-O-alkyl) comprises in total from 1 to 18 carbon atoms: such groups include methoxymethyl (—CH₂OCH₃), 2-methoxyethyl (—CH₂CH₂OCH₃) and 2-ethoxyethyl.

The term “C₂-C₄ alkylene” as used herein, is defined as saturated, divalent hydrocarbon radical having from 2 to 4 carbon atoms.

The term “C₃-C₃₀ cycloalkyl” is understood to mean an optionally substituted, saturated, mono-, bi- or tricyclic hydrocarbon group having from 3 to 30 carbon atoms. In general, a preference for cycloalkyl groups containing from 3-18 carbon atoms (C₃-C₁₈ cycloalkyl groups) should be noted. Examples of cycloalkyl groups include: cyclopropyl; cyclobutyl; cyclopentyl; cyclohexyl; cycloheptyl; cyclooctyl; adamantane; and, norbornane.

As used herein, an “C₆-C₁₈ aryl” group used alone or as part of a larger moiety—as in “aralkyl group”—refers to optionally substituted, monocyclic, bicyclic and tricyclic ring systems in which the monocyclic ring system is aromatic or at least one of the rings in a bicyclic or tricyclic ring system is aromatic. The bicyclic and tricyclic ring systems include benzofused 2-3 membered carbocyclic rings. Exemplary aryl groups include: phenyl; (C₁-C₄)alkylphenyl, such as tolyl and ethylphenyl; indenyl; naphthalenyl, tetrahydronaphthyl, tetrahydroindenyl; tetrahydroanthracenyl; and, anthracenyl. And a preference for phenyl groups may be noted.

As used herein, “C₂-C₂₀ alkenyl” refers to hydrocarbon groups having from 2 to 20 carbon atoms and at least one unit of ethylenic unsaturation. The alkenyl group can be straight chained, branched or cyclic and may optionally be substituted. The term “alkenyl” also encompasses radicals having “cis” and “trans” configurations, or alternatively, “E” and “Z” configurations, as appreciated by those of ordinary skill in the art. In general, however, a preference for unsubstituted alkenyl groups containing from 2 to 10 (C₂₋₁₀) or 2 to 8 (C₂₋₈) carbon atoms should be noted. Examples of said C₂-C₁₂ alkenyl groups include, but are not limited to:

—CH=CH₂; —CH=CHCH₃; —CH₂CH=CH₂;
 —C(=CH₂)(CH₃); —CH=CHCH₂CH₃;
 —CH₂CH=CHCH₃; —CH₂CH₂CH=CH₂; —CH=C(CH₃)₂; —CH₂C(=CH₂)(CH₃); —C(=CH₂)CH₂CH₃;
 —C(CH₃)=CHCH₃; —C(CH₃)CH=CH₂;
 —CH=CHCH₂CH₂CH₃; —CH₂CH=CHCH₂CH₃;
 —CH₂CH₂CH=CHCH₃; —CH₂CH₂CH₂CH=CH₂;
 —C(=CH₂)CH₂CH₂CH₃; —C(CH₃)=CHCH₂CH₃; —CH(CH₃)CH=CHCH₃; —CH(CH₃)CH₂CH=CH₂;
 —CH₂CH=C(CH₃)₂; 1-cyclopent-1-enyl; 1-cyclopent-2-enyl; 1-cyclopent-3-enyl; 1-cyclohex-1-enyl; 1-cyclohex-2-enyl; and, 1-cyclohexyl-3-enyl.

As used herein, “alkylaryl” refers to alkyl-substituted aryl groups and “substituted alkylaryl” refers to alkylaryl groups further bearing one or more substituents as set forth above. Further, as used herein “aralkyl” means an alkyl group substituted with an aryl radical as defined above.

The term “hetero” as used herein refers to groups or moieties containing one or more heteroatoms, such as N, O, Si and S. Thus, for example “heterocyclic” refers to cyclic groups having, for example, N, O, Si or S as part of the ring structure. “Heteroalkyl”, “heterocycloalkyl” and “heteroaryl” moieties are alkyl, cycloalkyl and aryl groups as defined hereinabove, respectively, containing N, O, Si or S as part of their structure.

As used herein, the term “epoxide” denotes a compound characterized by the presence of at least one cyclic ether group, namely one wherein an ether oxygen atom is attached to two adjacent carbon atoms thereby forming a cyclic structure. The term is intended to encompass monoepoxide compounds, polyepoxide compounds (having two or more epoxide groups) and epoxide terminated prepolymers. The term “monoepoxide compound” is meant to denote epoxide

compounds having one epoxy group. The term “polyepoxide compound” is meant to denote epoxide compounds having at least two epoxy groups. The term “diepoxide compound” is meant to denote epoxide compounds having two epoxy groups.

The epoxide may be unsubstituted but may also be inertly substituted. Exemplary inert substituents include chlorine, bromine, fluorine and phenyl.

As employed herein a “primary amino group” refers to an NH_2 group that is attached to an organic radical, and a “secondary amino group” refers to an NH group that is attached to two organic radicals, which may also together be part of a ring. The term “tertiary amine” thus references a nitrogen bearing moiety of which a nitrogen atom is not bound to a hydrogen atom. Where used, the term “amine hydrogen” refers to the hydrogen atoms of primary and secondary amino groups.

The term “equivalent weight” as used herein refers to the molecular weight divided by the number of a function concerned. As such, “epoxy equivalent weight” (EEW) means the weight of resin, in grams, that contains one equivalent of epoxy.

The “amine equivalent weight” is a calculated value (g/eq.) determined from the amine number. That amine number is determined by titration of the amine acetate ion by a dilute, typically 1 N HCl solution. For a pure material, the amine number can be calculated using the molecular weights of the pure compound and KOH (56.1 g/mol).

The term “Lewis acid” used herein denotes any molecule or ion—often referred to as an electrophile—capable of combining with another molecule or ion by forming a covalent bond with two electrons from the second molecule or ion: a Lewis acid is thus an electron acceptor.

The molecular weights referred to in this specification can be measured with gel permeation chromatography (GPC) using polystyrene calibration standards, such as is done according to ASTM 3536.

Viscosities of the compositions described herein are, unless otherwise stipulated, measured using the Anton Paar Viscometer, Model MCR 301 at standard conditions of 25° C. and 50% Relative Humidity (RH). The viscometer is calibrated one time a year and checked by services. The calibration is done with using special oil of known viscosity, which vary from 5,000 cps to 50,000 cps (parallel plate PP25 and at shear rate 1 1/s at 23° C.). Measurements of the compositions according to the present invention are done using the parallel plate PP20 at different shear rates from 1.5 1/s to 100 1/s.

DETAILED DESCRIPTION OF THE INVENTION

First Part of the Two-Part (2K) Composition

The first part of the two-part (2K) composition comprises: epoxy resin; and, electrolyte; solubilizer; and, filler.

Epoxy Resin

The first part of the composition comprises epoxy resin which will typically be present in an amount of from 30 to 70 wt. %, based on the weight of said composition: it is preferred that the epoxy resin constitutes from 35 to 60 wt. %, for example from 40 to 60 wt. % or from 47 to 57 wt. % of said composition.

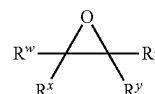
These epoxy resin quantities are preferred because quantity greater than 70% may adversely affect lap shear strength and debonding effect, whereas low quantities, mainly below 30% may lead to decrease in adhesive properties.

Epoxy resins as used herein may include mono-functional epoxy resins, multi- or poly-functional epoxy resins, and combinations thereof. The epoxy resins may be pure compounds but equally may be mixtures epoxy functional compounds, including mixtures of compounds having different numbers of epoxy groups per molecule. An epoxy resin may be saturated or unsaturated, aliphatic, cycloaliphatic, aromatic or heterocyclic and may be substituted. Further, the epoxy resin may also be monomeric or polymeric.

Without intention to limit the present invention, illustrative monoepoxide compounds include: alkylene oxides; epoxy-substituted cycloaliphatic hydrocarbons, such as cyclohexene oxide, vinylcyclohexene monoxide, (+)-cis-limonene oxide, (+)-cis,trans-limonene oxide, (–)-cis,trans-limonene oxide, cyclooctene oxide, cyclododecene oxide and α -pinene oxide; epoxy-substituted aromatic hydrocarbons; monoepoxy substituted alkyl ethers of monohydric alcohols or phenols, such as the glycidyl ethers of aliphatic, cycloaliphatic and aromatic alcohols; monoepoxy-substituted alkyl esters of monocarboxylic acids, such as glycidyl esters of aliphatic, cycloaliphatic and aromatic monocarboxylic acids; monoepoxy-substituted alkyl esters of polycarboxylic acids wherein the other carboxy group(s) are esterified with alkanols; alkyl and alkenyl esters of epoxy-substituted monocarboxylic acids; epoxyalkyl ethers of polyhydric alcohols wherein the other OH group(s) are esterified or etherified with carboxylic acids or alcohols; and, monoesters of polyhydric alcohols and epoxy monocarboxylic acids, wherein the other OH group(s) are esterified or etherified with carboxylic acids or alcohols.

By way of example, the following glycidyl ethers might be mentioned as being particularly suitable monoepoxide compounds for use herein: methyl glycidyl ether; ethyl glycidyl ether; propyl glycidyl ether; butyl glycidyl ether; pentyl glycidyl ether; hexyl glycidyl ether; cyclohexyl glycidyl ether; octyl glycidyl ether; 2-ethylhexyl glycidyl ether; allyl glycidyl ether; benzyl glycidyl ether; phenyl glycidyl ether; 4-tert-butylphenyl glycidyl ether; 1-naphthyl glycidyl ether; 2-naphthyl glycidyl ether; 2-chlorophenyl glycidyl ether; 4-chlorophenyl glycidyl ether; 4-bromophenyl glycidyl ether; 2,4,6-trichlorophenyl glycidyl ether; 2,4,6-tribromophenyl glycidyl ether; pentafluorophenyl glycidyl ether; o-cresyl glycidyl ether; m-cresyl glycidyl ether; and, p-cresyl glycidyl ether.

In an embodiment, the monoepoxide compound conforms to Formula (1) herein below:



wherein: R^w , R^x , R^y and R^z may be the same or different and are independently selected from hydrogen, a halogen atom, a C_1 - C_8 alkyl group, a C_3 to C_{10} cycloalkyl group, a C_2 - C_{12} alkenyl, a C_6 - C_{18} aryl group or a C_7 - C_{18} aralkyl group, with the proviso that at least one of R^y and R^z is not hydrogen.

It is preferred that R^w , R^x and R^y are hydrogen and R^z is either a phenyl group or a C_1 - C_8 alkyl group and, more preferably, a C_1 - C_4 alkyl group. Having regard to this embodiment, exemplary monoepoxides include: ethylene oxide; 1,2-propylene oxide (propylene oxide); 1,2-butylene oxide; cis-2,3-epoxybutane; trans-2,3-epoxybutane; 1,2-ep-

oxypentane; 1,2-epoxyhexane; 1,2-heptylene oxide; decene oxide; butadiene oxide; isoprene oxide; and, styrene oxide.

In the present invention, reference is made to using at least one monoepoxide compound selected from the group consisting of: ethylene oxide; propylene oxide; cyclohexene oxide; (+)-cis-limonene oxide; (+)-cis,trans-limonene oxide; (-)-cis,trans-limonene oxide; cyclooctene oxide; and, cyclododecene oxide.

Again, without intention to limit the present invention, suitable polyepoxide compounds may be liquid, solid or in solution in solvent. Further, such polyepoxide compounds should have an epoxide equivalent weight of from 100 to 700 g/eq, for example from 120 to 320 g/eq. And generally, diepoxide compounds having epoxide equivalent weights of less than 500 g/eq, or even less than 400 g/eq, are preferred: this is predominantly from a cost standpoint, as in their production, lower molecular weight epoxy resins require more limited processing in purification.

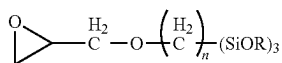
As examples of types or groups of polyepoxide compounds which may be polymerized in present invention, mention may be made of: glycidyl ethers of polyhydric alcohols and polyhydric phenols; glycidyl esters of polycarboxylic acids; and, epoxidized polyethylenically unsaturated hydrocarbons, esters, ethers and amides.

Suitable diglycidyl ether compounds may be aromatic, aliphatic or cycloaliphatic in nature and, as such, can be derivable from dihydric phenols and dihydric alcohols. And useful classes of such diglycidyl ethers are: diglycidyl ethers of aliphatic and cycloaliphatic diols, such as 1,2-ethanediol, 1,4-butanediol, 1,6-hexanediol, 1,8-octanediol, 1,12-dodecanediol, cyclopentane diol and cyclohexane diol; bisphenol A based diglycidylethers; bisphenol F diglycidyl ethers; diglycidyl o-phthalate, diglycidyl isophthalate and diglycidyl terephthalate; polyalkyleneglycol based diglycidyl ethers, in particular polypropyleneglycol diglycidyl ethers; and, polycarbonatediol based glycidyl ethers. Other suitable diepoxides which might also be mentioned include: diepoxides of double unsaturated fatty acid C₁-C₁₈ alkyl esters; butadiene diepoxide; polybutadiene diglycidyl ether; vinyl-cyclohexene diepoxide; and, limonene diepoxide.

Further illustrative polyepoxide compounds include but are not limited to: glycerol polyglycidyl ether; trimethylolpropane polyglycidyl ether; pentaerythritol polyglycidyl ether; diglycerol polyglycidyl ether; polyglycerol polyglycidyl ether; and, sorbitol polyglycidyl ether.

And examples of highly preferred polyepoxide compounds include: bisphenol-A epoxy resins, such as DERTM 331, DERTM 330 and DERTM 383; bisphenol-F epoxy resins, such as DERTM 354; bisphenol-A/F epoxy resin blends, such as DERTM 353; aliphatic glycidyl ethers, such as DERTM 736; polypropylene glycol diglycidyl ethers, such as DERTM 732; solid bisphenol-A epoxy resins, such as DERTM 661 and DERTM 664 UE; solutions of bisphenol-A solid epoxy resins, such as DERTM 671-X75; epoxy novolac resins, such as DENTM 438; brominated epoxy resins such as DERTM 542; castor oil triglycidyl ether, such as ERISYSTM GE-35H; polyglycerol-3-polyglycidyl ether, such as ERISYSTM GE-38; and, sorbitol glycidyl ether, such as ERISYSTM GE-60.

The above aside, the first part of the composition can in certain embodiments comprise glycidoxo alkyl alkoxy silanes having the formula:



wherein: each R is independently selected from methyl or ethyl; and, n is from 1-10.

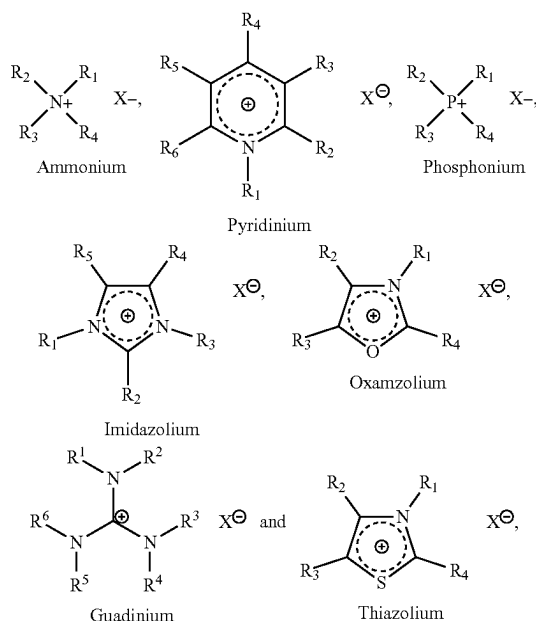
Exemplary silanes include but are not limited to: γ -glycidoxo propyl trimethoxy silane, γ -glycidoxo ethyl trimethoxy silane, γ -glycidoxo methyl trimethoxy silane, γ -glycidoxo methyl triethoxy silane, γ -glycidoxo ethyl triethoxy silane, γ -glycidoxo propyl triethoxy silane; and, 8-glycidooxyoctyl trimethoxysilane. When present, the epoxide functional silanes should constitute less than less than 20 wt. %, preferably less than 10 wt. % or less than 5 wt. %, based on the total weight of the epoxide compounds.

The present invention also does not preclude the first part of the curable compositions from further comprising one or more cyclic monomers selected from the group consisting of: oxetanes; cyclic carbonates; cyclic anhydrides; and, lactones. The disclosures of the following citations may be instructive in disclosing suitable cyclic carbonate functional compounds: U.S. Pat. Nos. 3,535,342; 4,835,289; 4,892,954; UK Patent No. GB-A-1,485,925; and, EP-A-0 119 840. However, such cyclic co-monomers should constitute less than 20 wt. %, preferably less than 10 wt. % or less than 5 wt. %, based on the total weight of the epoxide compounds. Electrolyte

The first part of the composition comprises from 2 to 25 wt. %, based on the weight of the composition, of electrolyte: the electrolyte may preferably constitute from 4 to 23 wt. %, for example from 5 to 20 wt. % of said composition.

These electrolyte quantities are preferred because quantity greater than 25% may lead to a good debonding effect, however the cure may be incomplete and therefore, initial adhesive properties may be adversely affected, whereas low quantities, mainly below 2%, may lead to a lack of debonding effect.

The electrolyte preferably comprises at least one salt having a Formula selected from the group consisting of:



wherein: R¹, R², R³, R⁴, R⁵ and R⁶ are independently selected from hydrogen, C₁-C₁₈ alkyl, C₃-C₁₈ cycloalkyl, C₆-C₁₈ aryl, C₇-C₂₄ aralkyl, C₂-C₂₀ alkenyl, -C(O)R^q, -C(O)OH, -CN and -NO₂; and, R^q is C₁-C₆ alkyl.

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For completeness, the terms C_1 - C_{18} alkyl, C_3 - C_{18} cycloalkyl, C_6 - C_{18} aryl, C_7 - C_{24} aralkyl, C_2 - C_{20} alkenyl expressly includes groups wherein one or more hydrogen atoms are substituted by halogen atoms (e.g. C_1 - C_{18} haloalkyl) or hydroxyl groups (e.g. C_1 - C_{18} hydroxyalkyl). In particular, it is preferred that R^1 , R^2 , R^3 , R^4 , R^5 and R^6 are independently selected from hydrogen, C_1 - C_{12} alkyl, C_1 - C_{12} haloalkyl, C_1 - C_{12} hydroxyalkyl and C_3 - C_{12} cycloalkyl. For example, R^1 , R^2 , R^3 , R^4 , R^5 and R^6 may be independently selected from hydrogen, C_1 - C_6 alkyl, C_1 - C_6 haloalkyl and C_1 - C_6 hydroxyalkyl.

There is no particular intention to limit the counter anion (X^-) which may be employed in the electrolyte. Exemplary anions may be selected from:

Halides;

Pseudohalides and halogen-containing compounds of the formulae PF_6^- , $CF_3SO_3^-$, $(CF_3SO_3)_2N^-$, $CF_3CO_2^-$ and $CCl_3CO_2^-$;

CN^- , SCN^- and OCN^- ;

Phenates;

Sulfates, sulfites and sulfonates of the general formulae SO_4^{2-} , HSO_4^- , SO_3^{2-} , HSO_3^- , $R^aOSO_3^-$ and $R^aSO_3^-$;

Phosphates of the general formulae PO_4^{3-} , HPO_4^{2-} , $H_2PO_4^-$, $R^aPO_4^{2-}$, $HR^aPO_4^-$ and $R^aR^bPO_4^-$;

Phosphonates and phosphinates of the general formulae $R^aHPO_3^-$, $R^aR^bPO_2^-$ and $R^aR^bPO_3^-$;

Phosphites of the general formulae: PO_3^{3-} , HPO_3^{2-} , $H_2PO_3^-$, $R^aPO_3^{2-}$, $R^aHPO_3^-$ and $R^aR^bPO_3^-$;

Phosphonites and phosphinites of the general formulae $R^aR^bPO_2^-$, $R^aHPO_2^-$, $R^aR^bPO^-$ and R^aHPO^- ;

Carboxylic acid anions of the general formula R^aCOO^- ;

Hydroxycarboxylic acids anions and sugar acid anions;

Saccharinates (salts of o-benzoic acid sulfimide);

Borates of the general formulae BO_3^{3-} , HBO_3^{2-} , $H_2BO_3^-$, $R^aR^bBO_3^-$, $R^aHBO_3^{2-}$, $R^aBO_3^{2-}$, $B(OR^a)(OR^b)(OR^c)(OR^d)^-$, $B(HSO_4)$ and $B(R^aSO_4)^-$;

Boronates of the general formulae $R^aBO_2^{2-}$ and $R^aR^bBO^-$;

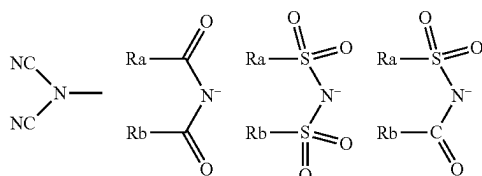
Carbonates and carbonic acid esters of the general formulae HCO_3^- , CO_3^{2-} and $R^aCO_3^-$;

Silicates and silicic acid esters of the general formulae SiO_4^{4-} , $HSiO_4^{3-}$, $H_2SiO_4^{2-}$, $H_3SiO_4^-$, $R^aSiO_4^{3-}$, $R^aR^bSiO_4^{2-}$, $R^aR^bR^cSiO_4^-$, $HR^aSiO_4^{2-}$, $H_2R^aSiO_4^{2-}$ and $HR^aR^bSiO_4^-$;

Alkyl- and arylsilanates of the general formulae $R^aSiO_3^{3-}$, $R^aR^bSiO_2^{2-}$, $R^aR^bR^cSiO^-$, $R^aR^bR^cSiO_3^-$, $R^aR^bR^cSiO_2^-$ and $R^aR^bSiO_3^{2-}$;

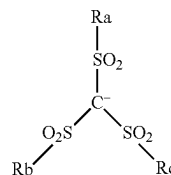
Pyridinates and pyrimidinates;

Carboxylic acid imides, bis(sulfonyl)imides and sulfonylimides of the general formulae:



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Methides of the general formula:



Alkoxides and aryloxides of the general formula R^aO^- ; and,

Sulfides, hydrogen sulfides, polysulfides, hydrogen polysulfides and thiolates of the general formulae S^{2-} , HS^- , $[S_n]^{2-}$, $[HS_n]^-$ and $[R^aS_n]^-$ in which general formulae v is a whole positive number of from 2 to 10.

R^a , R^b , R^c and R^d are independently selected from hydrogen, a C_1 - C_{12} alkyl, C_5 - C_{12} cycloalkyl, C_5 - C_{12} heterocycloalkyl, C_6 - C_{18} aryl and C_5 - C_{18} heteroaryl.

Based on the definitions in the above list, preferred anions are selected from the group consisting of: halides; pseudohalides and halogen-containing compounds as defined above; carboxylic acid anions, in particular formate, acetate, propionate, butyrate and lactate; hydroxycarboxylic acid anions; pyridinates and pyrimidinates; carboxylic acid imides, bis(sulfonyl)imides and sulfonylimides; sulfates, in particular methyl sulfate and ethyl sulfate; sulfites; sulfonates, in particular methansulfonate; and, phosphates, in particular dimethyl-phosphate, diethyl-phosphate and di-(2-ethylhexyl)-phosphate.

The electrolyte of the first part is preferably selected from the group consisting of 1-ethyl-3-methylimidazolium methansulfonate, 1-ethyl-3-methylimidazolium methyl sulfate, 1-hexyl-3-methylimidazolium 2-(2-fluoroanilino)-pyridinate, 1-hexyl-3-methylimidazolium imide, 1-butyl-1-methyl-pyrrolidinium 2-(2-fluoroanilino)-pyridinate, 1-butyl-1-methyl-pyrrolidinium imide, trihexyl(tetradecyl) phosphonium 2-(2-fluoroanilino)-pyridinate, cyclohexyltrimethylammonium bis(trifluoromethylsulfonyl) imide, di(2-hydroxyethyl) ammonium trifluoroacetate, N,N-dimethyl (2-hydroxyethyl) ammonium octanoate, methyltriocetylammmonium bis(trifluoromethylsulfonyl) imide, N-ethyl-N—N—N-tetramethylguanidinium trifluoromethanesulfonate, guanidinium trifluoromethanesulfonate, 1-butyl-4-methylpyridinium bromide, 1-butyl-3-methylpyridinium tetrafluoroborate, 1-butyl-3-hydroxymethylpyridinium ethylsulfate, 1-butyl-1-methylpyrrolidinium bis(trifluoromethylsulfonyl) imide, 1-butyl-methylpyrrolidinium tris(pentafluoroethyl) trifluorophosphate, 3-methyl imidazolium ethylsulfate, 1-ethyl-3-methylimidazolium chloride, 1-ethyl-3-ethyl-methylimidazolium bromide, 1-butyl-3-methylimidazolium chloride, 1-hexyl-3-methylimidazolium chloride, 1-octyl-3-methylimidazolium chloride, 1-methyl-3-octylimidazolium chloride, 1-propyl-3-methylimidazolium iodide, 1-butyl-3-methylimidazolium tetrafluoroborate, 1-butyl-3-methylimidazolium trifluoromethanesulfonate, 1-butyl-3-methylimidazolium hexafluorophosphate, 1-butyl-2,3-dimethylimidazolium tetrafluoroborate, 1-butyl-2,3-dimethylimidazolium hexafluorophosphate, 1-butylimidazol, 1-methylimidazolium tetrafluoroborate, tetrabutylphosphonium tris(pentafluoroethyl) trifluorophosphate, trihexyl(tetradecyl) phosphonium tetrafluoroborate and mixtures thereof. A particular preference for the use of at least one of 1-ethyl-3-methylimidazolium methansulfonate, 1-ethyl-3-methylimidazolium methyl sulfate may be mentioned.

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Above listed electrolytes are preferred because they provide good stability for the composition and provide good initial adhesion strength, and upon treatment with potential they provide good debonding effect.

Solubilizer

The first part of the two-part (2K) composition may comprise a solubilizer which will typically be present in an amount of from 0.5 to 15 wt. %, based on the weight of the composition; preferably the solubilizer constitutes from 1.5 to 10 wt. %, for example from 3 to 10 wt. % of said composition.

These solubilizer quantities are preferred because quantity greater than 15% may adversely affect adhesion and cure properties, whereas low quantities, mainly below 0.5% may lead the composition being more viscous/solid, and therefore, hinder blending of the ingredients.

The solubilizer has the function of promoting the miscibility of the electrolyte within the adhesive composition formed upon admixture of the two parts thereof: the solubilizer may or may not form part of the polymer matrix formed upon curing of the adhesive composition but does serve to facilitate ion transfer therein. The solubilizer is, as such, preferably a polar compound and should desirably be liquid at room temperature.

Suitable classes of solubilizer include: polyphosphazenes; polymethylenesulfides; polyoxyalkylene glycols; polyethylene imines; silicone surfactants and fluorinated silicone surfactants, such as fluorinated polysilanes, polyalkylsiloxane and polyoxyalkylene modified polydimethylsiloxanes; co-polymers of functionalized polyalkylsiloxanes and epoxy resins, such as copolymers of polydimethylsiloxane (PDMS) and epoxy resin; polypolyhydric alcohols; and, sugars.

Polyhydric alcohols and sugars such as ethylene glycol, 1,3-propanediol, cyclohexandiol, hydroquinone, catechol, resorcinol, phloroglucinol, pyrogallol, hydroxyhydroquinone, tris(hydroxymethyl)benzene, tris(hydroxymethyl)benzene with three methyl or ethyl substituents bonded to the remaining benzene carbon atoms, isosorbide, isomannide, isoidide, glycerol, cyclohexane-1,2,4-triol, 1,3,5-cyclohexanetriol, pentane-1,2,3-triol, hexane-1,3,5-triol, erythritol, 1,2,4,5-tetrahydroxybenzene, threitol, arabitol, xylitol, ribitol, mannitol, sorbitol, inositol, fructose, glucose, mannose, lactose, 1,1,1-tris(hydroxymethyl)propane, 1,1,1-tris(hydroxymethyl)ethane, di(trimethylolpropane), trimethylolpropane ethoxylate, 2-hydroxymethyl-1,3-propanediol, pentaerythritol allyl ether and pentaerythritol.

Of the polyoxyalkylene glycols, a particular preference for the use of polyoxy(C₂-C₃)alkylene glycols having a weight average molecular weight of from 200 to 10000 g/mol, for example 200 to 2000 g/mol, may be noted.

Second Part of the Two-Part (2K) Composition

The second part of the two-part composition necessarily comprises: a curing agent; and, an accelerator.

Curing Agent

The curative necessarily consists of at least one compound possessing at least two epoxide reactive groups per molecule. Curatives may, in particular, include one or more of: i) at least one polyamine having at least two amine hydrogens reactive toward epoxide groups; ii) at least one mercapto compound having at least two mercapto groups reactive toward epoxide groups; and, iii) at least one Mannich base.

The at least one polyamine having at least two amine hydrogens reactive toward epoxide groups should, in particular, contain primary and/or secondary amine groups and

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have an equivalent weight per primary or secondary amine group of not more than 150 g/eq., more preferably not more than 125 g/eq.

Suitable polyamines, which may be used alone or in combination, include but are not limited to the following:

i) Aliphatic, cycloaliphatic or arylaliphatic primary diamines of which the following examples may be mentioned: 2,2-dimethyl-1,3-propanediamine; 1,3-pentanediamine (DAMP); 1,5-pentanediamine; 1,5-diamino-2-methylpentane (MPMD); 2-butyl-2-ethyl-1,5-pentanediamine (C11-neodiamine); 1,6-hexanediamine (hexamethylenediamine, HMDA); 2,5-dimethyl-1,6-hexanediamine; 2,2,4- and/or 2,4,4-trimethylhexamethylenediamine; 1,7-heptanediamine; 1,8-octanediamine; 1,9-nonanediamine; 1,10-decanediamine; 1,11-undecanediamine; 1,12-dodecanediamine; 1,2-, 1,3- and 1,4-diaminocyclohexane; bis(4-aminocyclohexyl)methane; bis(4-amino-3-methylcyclohexyl)methane; bis(4-amino-3-ethylcyclohexyl)methane; bis(4-amino-3,5-dimethylcyclohexyl)methane; bis(4-amino-3-ethyl-5-methylcyclohexyl)methane; 1-amino-3-aminomethyl-3,5,5-trimethylcyclohexane (isophorone diamine, IPDA); 2- and/or 4-methyl-1,3-diaminocyclohexane; 1,3-bis(aminomethyl)-cyclohexane; 1,4-bis(aminomethyl)cyclohexane; 2,5(2,6)-bis(aminomethyl)-bicyclo[2.2.1]heptane (norborane diamine, NBDA); 3(4),8(9)-bis(aminomethyl)tricyclo[5.2.1.0(2,6)]-decane (TCD-diamine); 1,4-diamino-2,2,6-trimethylcyclohexane (TMCD); 1,8-menthanediamine; 3,9-bis(3-aminopropyl)-2,4,8,10-tetraoxaspiro[5.5]undecane; and, 1,3-bis(aminomethyl)benzene (MXDA).

ii) Tertiary amine group-containing polyamines with two or three primary aliphatic amine groups of which the following specific examples may be mentioned: N,N'-bis(aminopropyl)-piperazine; N,N-bis(3-aminopropyl)methylamine; N,N-bis(3-aminopropyl)ethylamine; N,N-bis(3-aminopropyl)propylamine; N,N-bis(3-aminopropyl)cyclohexylamine; N,N-bis(3-aminopropyl)-2-ethyl-hexylamine; tris(2-aminoethyl)amine; tris(2-aminopropyl)amine; tris(3-aminopropyl)amine; and, the products from the double cyanoethylation and subsequent reduction of fatty amines derived from natural fatty acids, such as N,N-bis(3-aminopropyl)dodecylamine and N,N-bis(3-aminopropyl)tallow alkylamine, commercially available as Triameen® Y12D and Triameen® YT (from Akzo Nobel).

iii) Ether group-containing aliphatic primary polyamines of which the following specific examples may be mentioned: bis(2-aminoethyl)ether; 3,6-dioxaoctane-1,8-diamine; 4,7-dioxadecane-1,10-diamine; 4,7-dioxadecane-2,9-diamine; 4,9-dioxadodecane-1,12-diamine; 5,8-dioxadodecane-3,10-diamine; 4,7,10-trioxatridecane-1,13-diamine and higher oligomers of these diamines; bis(3-aminopropyl)polytetrahydrofuranes and other polytetrahydrofuran diamines; cycloaliphatic ether group-containing diamines obtained from the propoxylation and subsequent amination of 1,4-dimethylcyclohexane, such as that material commercially available as Jeffamine® RFD-270 (from Huntsman); polyoxyalkylenedi- or -triamines obtainable as products from the amination of polyoxyalkylenedi- and -triols and which are commercially available under the name of Jeffamine® (from Huntsman), under the name of polyetheramine (from BASF) or under the name of PC Amines® (from Nitroil). A particular preference may be noted for the use of Jeffamine® D-230, Jef-

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famine® D-400, Jeffamine® D-600, Jeffamine® D-2000, Jeffamine® D-4000, Jeffamine® T-403, Jeffamine® T-3000, Jeffamine® T-5000, Jeffamine® EDR-104, Jeffamine® EDR-148 and Jeffamine® EDR-176, as well as corresponding amines from BASF or Nitroil.

iv) Primary diamines with secondary amine groups of which the following examples may be mentioned: 3-(2-aminoethyl)aminopropylamine, bis(hexamethylene)triamine (BHMT); diethylenetriamine (DETA); triethylenetetramine (TETA); tetraethylenepentamine (TEPA); pentaethylenhexamine (PEHA); higher homologs of linear polyethyleneamines, such as polyethylene polyamines with 5 to 7 ethyleneamine units (so-called "higher ethylenepolyamine," HEPA); products from the multiple cyanoethylation or cyanobutyl-
 ation and subsequent hydrogenation of primary di- and polyamines with at least two primary amine groups, such as dipropylenetriamine (DPTA), N-(2-aminoethyl)-1,3-propanediamine (N3-amine), N,N'-bis(3-aminopropyl)ethylenediamine (N4-amine), N,N'-bis(3-aminopropyl)-1,4-diaminobutane, N5-(3-aminopropyl)-2-methyl-1,5-pentanediamine, N3-(3-aminopentyl)-1,3-pentanediamine, N5-(3-amino-1-ethylpropyl)-2-methyl-1,5-pentanediamine or N,N'-bis(3-amino-1-ethylpropyl)-2-methyl-1,5-pentanediamine.

v) Polyamines with one primary and at least one secondary amino group of which the following examples may be mentioned: N-butyl-1,2-ethanediamine; N-hexyl-1,2-ethanediamine; N-(2-ethylhexyl)-1,2-ethanediamine; N-cyclohexyl-1,2-ethanediamine; 4-aminomethylpiperidine; N-(2-aminoethyl)piperazine; N-methyl-1,3-propanediamine; N-butyl-1,3-propanediamine; N-(2-ethylhexyl)-1,3-propanediamine; N-cyclohexyl-1,3-propanediamine; 3-methylamino-1-pentylamine; 3-ethylamino-1-pentylamine; 3-cyclohexylamino-1-pentylamine; fatty diamines such as N-cocoalkyl-1,3-propanediamine; products from the Michael-type addition reaction of primary aliphatic diamines with acrylonitrile, maleic or fumaric acid diesters, citraconic acid diesters, acrylic and methacrylic acid esters, acrylic and methacrylic acid amides and itaconic acid diesters, reacted in a 1:1 molar ratio; products from the partial reductive alkylation of primary polyamines with aldehydes or ketones, especially N-monoalkylation products of the previously mentioned polyamines with two primary amine groups and in particular of 1,6-hexanediamine, 1,5-diamino-2-methylpentane, 1,3-bis(aminomethyl)cyclohexane, 1,4-bis(aminomethyl)cyclohexane, 1,3-bis(aminomethyl)benzene, BHMT, DETA, TETA, TEPA, DPTA, N3-amine and N4-amine, wherein preferred alkyl groups are benzyl, isobutyl, hexyl and 2-ethylhexyl; and, partially styrenated polyamines such as those commercially available as Gaskamine® 240 (from Mitsubishi Gas Chemical).

vi) Secondary diamines and, in particular, N,N'-dialkylation products of the previously mentioned polyamines with two primary amine groups, especially N,N'-dialkylation products of 1,6-hexanediamine, 1,5-diamino-2-methylpentane, 1,3-bis(aminomethyl)cyclohexane, 1,4-bis(aminomethyl)cyclohexane, 1,3-bis(aminomethyl)benzene, BHMT, DETA, TETA, TEPA, DPTA, N3-amine or N4-amine, wherein preferred alkyl groups are 2-phenylethyl, benzyl, isobutyl, hexyl and 2-ethylhexyl.

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vii) Aromatic polyamines of which mention may be made of: m- and p-phenylenediamine; 4,4', 2,4' and 2,2'-diaminodiphenylmethane; 3,3'-dichloro-4,4'-diaminodiphenylmethane (MOCA); 2,4- and 2,6-tolylenediamine; mixtures of 3,5-dimethylthio-2,4- and -2,6-tolylenediamine (available as Ethacure® 300 from Albermarle); mixtures of 3,5-diethyl-2,4- and -2,6-tolylene diamine (DETDA); 3,3',5,5'-tetraethyl-4,4'-diaminodiphenylmethane (M-DEA); 3,3',5,5'-tetraethyl-2,2'-dichloro-4,4'-diaminodiphenylmethane (M-CDEA); 3,3'-diisopropyl-5,5'-dimethyl-4,4'-diaminodiphenylmethane (M-MIPA); 3,3',5,5'-tetraisopropyl-4,4'-diaminodiphenylmethane (M-DIPA); 4,4'-diaminodiphenyl-sulfone (DDS); 4-amino-N-(4-aminophenyl)benzenesulfonamide; 5,5'-methylene-dianthranilic acid; dimethyl-(5,5'-methylene-dianthranilate); 1,3-propylene-bis(4-aminobenzoate); 1,4-butylene-bis(4-aminobenzoate); polytetramethylene oxide-bis(4-aminobenzoate) (available as Versalink® from Air Products); 1,2-bis(2-aminophenylthio)ethane, 2-methylpropyl-(4-chloro-3,5-diaminobenzoate); and, tert.butyl-(4-chloro-3,5-diaminobenzoate).

viii) Polyamidoamines of which indicative members include the reaction products of monohydric or polyhydric carboxylic acids or the esters or anhydrides thereof, —in particular dimer fatty acids—and an aliphatic, cycloaliphatic or aromatic polyamine, for instance polyalkyleneamines such as DETA or TETA. Commercially available polyamidoamines include: Versamid® 100, 125, 140 and 150 (from Cognis); Aradur® 223, 250 and 848 (from Huntsman); Euretek® 3607 and 530 (from Huntsman); and, Beckopox® EH 651, EH 654, EH 655, EH 661 and EH 663 (from Cytec).

Preferred among the aforementioned polyamines having at least two primary aliphatic amine groups are: isophorone diamine (IPDA); hexamethylene diamine (HMDA); 1,3-bis(aminomethyl)cyclohexane; 1,4-bis(aminomethyl)cyclohexane; bis(4-amino-cyclohexyl)methane; bis(4-amino-3-methylcyclohexyl)methane; NBDA; and, ether group-containing polyamines with a number average molecular weight (Mn) of up to 500 g/mol. Particularly preferred among said ether group-containing polyamines are Jeffamine® D-230 and D-600 (available from Huntsman).

As noted above, the composition of the present invention may optionally comprise at least one compound which has at least two reactive mercapto-groups per molecule. Suitable mercapto-group containing compounds, which may be used alone or in combination, include but are not limited to the following.

Liquid mercaptan-terminated polysulfide polymers of which commercial examples include: Thiokol® polymers (available from Morton Thiokol), in particular the types LP-3, LP-33, LP-980, LP-23, LP-55, LP-56, LP-12, LP-31, LP-32 and LP-2 thereof; and, Thioplast® polymers (from Akzo Nobel), in particular the types G10, G112, G131, G1, G12, G21, G22, G44 and G4.

Mercaptan-terminated polyoxyalkylene ethers, obtainable by reacting polyoxyalkylenedi- and -triols either with epichlorohydrin or with an alkylene oxide, followed by sodium hydrogen sulfide.

Mercaptan-terminated compounds in the form of polyoxyalkylene derivatives, known under the trade name of Capcure® (from Cognis), in particular the types WR-8, LOF and 3-800 thereof.

Polyesters of thiocarboxylic acids of which particular examples include: pentaerythritol tetramercaptoacetate (PETMP); trimethylolpropane trimercaptoacetate (TMPMP); glycol dimercaptoacetate; and, the esterification products of polyoxyalkylene diols and triols, ethoxylated trimethylolpropane and polyester diols with thiocarboxylic acids such as thioglycolic acid and 2- or 3-mercaptopropionic acid.

2,4,6-trimercapto-1,3,5-triazine, 2,2'-(ethylenedioxy)-dithianethiol (triethylene glycol dimercaptan) and/or ethanedithiol.

A preference for the use of polyesters of thiocarboxylic acids and, in particular, for the use of at least one of pentaerythritol tetramercaptoacetate (PETMP), trimethylolpropane trimercaptoacetate (TMPMP) and glycol dimercaptoacetate is acknowledged.

As noted above, the curative may comprise at least one Mannich base. Such compounds may be characterized by containing at least one phenalkamine and, in particular, a phenalkamine obtained from the condensation of cardanol (CAS Number: 37330-39-5), an aldehyde and an amine. The reactant amine in the condensation reaction is desirably ethylenediamine or diethyltriamine.

Mannich bases and phenalkamines are known in the art and suitable examples include the commercially available phenalkamines Cardolite® NC-541, NC-557, NC-558, NC-566, Lite 2001 and Lite 2002 (available from Cardolite), Aradur® 3440, 3441, 3442 and 3460 (available from Huntsman) and Beckpox® EH 614, EH 621, EH 624, EH 628 and EH 629 (available from Cytec).

Accelerator

Suitable accelerators are substances that promote the reaction between the epoxide groups and the epoxide-reactive groups, for instance the reaction between the amine or thiol groups and the epoxide groups. A specific example relates to the use of an amine accelerator which functions by de-protonation of reactive thiol (—SH) groups present to thiolate (—S⁻), which thiolate reacts with epoxy group by nucleophilic ring opening polymerization.

Without intention to limit the accelerators used in the present invention, mention may be made of the following suitable accelerators: i) acids or compounds hydrolyzable to acids, in particular a) organic carboxylic acids, such as acetic acid, benzoic acid, salicylic acid, 2-nitrobenzoic acid and lactic acid; b) organic sulfonic acids, such as methanesulfonic acid, p-toluenesulfonic acid and 4-dodecylbenzenesulfonic acid; c) sulfonic acid esters; d) inorganic acids, such as phosphoric acid; e) Lewis acid compounds, such as BF₃ amine complexes, SbF₆ sulfonium compounds, bis-arene iron complexes; f) Bronsted acid compounds, such as pentafluoroantimonic acid complexes; and, e) mixtures of the aforementioned acids and acid esters; ii) tertiary amines, such as 2piperazin-1-ylethanamine, 1,4-diazabicyclo[2.2.2]octane, benzyldimethylamine, α -methylbenzyl dimethylamine, triethanolamine, dimethylamino propylamine, imidazoles—including N-methylimidazole, N-vinylimidazole and 1,2-dimethylimidazole—and salts of such tertiary amines; iii) quaternary ammonium salts, such as benzyldimethyl ammonium chloride; iv) amidines, such as 1,8-diazabicyclo[5.4.0]undec-7-ene; v) guanidines, such as 1,1,3,3-tetraethylguanidine; vi) phenols, in particular bisphenols; vii) phenol resins; and, viii) phosphites, such as di- and triphenylphosphites.

The skilled artisan will recognize that the selection of an accelerator is not simply concerned with adding the fastest accelerator. Other factors determinative in the selection of accelerators include: cost; toxicity; solubility; processing

effects, such as working time, premature gelation, exothermic degradation, expansion and off-gassing; final properties, such as glass transition temperature (T_g), modulus, strength, elongation at break and chemical resistance; regulatory concerns; and, ease of use.

It is preferred in the present invention to employ an accelerator comprising or consisting of at least one tertiary amine, at least one amidine or mixtures thereof. More particularly, accelerator should be selected from the group consisting of imidazole, methylimidazole, benzyldimethylamine, 1,8-diazabicyclo[5.4.0]undec-7-ene, 1,4-diazabicyclo[2.2.2]octane and mixtures thereof.

The accelerator should typically be present in the composition in an amount of from 0.1 to 15 wt. %, based on the weight of the composition: the accelerator may preferably constitute from 1 to 10 wt. %, for example from 5 to 10 wt. % of said composition.

These accelerator quantities are preferred because quantity greater than 15% may lead to an excess of accelerator in a composition, which may adversely affect cure process and adhesive properties, whereas low quantities, mainly below 0.1%, may lead no physical effect.

Electrically Non-Conductive Fillers

The two-part (2K) composition of the present invention is characterized by the presence of electrically non-conductive fillers. For completeness, these fillers may be disposed in one or both parts of the composition. The composition of the electrically non-conductive filler in each part is independently determined and, as such, may be the same or different for each part.

Broadly, there is no particular intention to limit the shape of the particles employed as non-conductive fillers: particles that are acicular, spherical, ellipsoidal, cylindrical, bead-like, cubic or platelet-like may be used alone or in combination. Moreover, it is envisaged that agglomerates of more than one particle type may be used. Equally, there is no particular intention to limit the size of the particles employed as non-conductive fillers. However, such non-conductive fillers will conventionally have an average volume particle size, as measured by laser diffraction/scattering methods, of from 0.1 to 1500 μ m, for example from 1 to 1000 μ m or from 1 to 500 μ m.

Exemplary non-conductive fillers include but are not limited to calcium carbonate, calcium oxide, calcium hydroxide (lime powder), precipitated and/or pyrogenic silicic acid, zeolites, bentonites, wollastonite, magnesium carbonate, diatomite, barium sulfate, alumina, clay, talc, sand, quartz, flint, mica, glass beads, glass powder, fumed silica, and other ground mineral substances. Short fibers such as glass fibers, glass filament, polyacrylonitrile, carbon fibers, or polyethylene fibers can also be added.

The pyrogenic and/or precipitated silicic acids advantageously have a BET surface area from 10 to 90 m²/g. When they are used, they do not cause any additional increase in the viscosity of the composition according to the present invention but do contribute to strengthening the cured composition.

Hydrophobicity may be required by the electrolyte for it to be dissolved into the composition and to avoid phase separation. Non-conductive fillers, and especially fumed silica, provide required hydrophobicity to the composition, and therefore, improve solubility of the electrolyte and prevent phase separation. Further, the technical data indicates that the presence of a non-electrically conductive filler improves the initial adhesive properties of the composition.

It is likewise conceivable to use pyrogenic and/or precipitated silicic acids having a higher BET surface area,

advantageously from 100 to 250 m²/g as a filler: because of the greater BET surface area, the effect of strengthening the cured composition is achieved with a smaller proportion by weight of silicic acid.

Also suitable as non-conductive fillers are hollow spheres having a mineral shell or a plastic shell. These can be, for example, hollow glass spheres that are obtainable commercially under the trade names Glass Bubbles®. Plastic-based hollow spheres, such as Expancel® or Dualite®, may be used and are described in EP 0 520 426 B1: they are made up of inorganic or organic substances and each have a diameter of 1 mm or less, preferably 500 µm or less preferably from 100 to 200 µm.

Non-conductive fillers which impart thixotropy to the composition may be preferred for many applications: such fillers are also described as rheological adjuvants, e.g. hydrogenated castor oil, fatty acid amides, or swellable plastics such as PVC.

In a highly preferred embodiment, the composition according to the present invention comprises fumed silica as non-electrically conductive filler.

The desired viscosities of each part of the composition independently and also the desired viscosity of the curable composition formed upon bringing the two parts together may be determinative of the amount of filler used. Having regard to that latter consideration, the total amount of fillers—both electrically conductive and non-conductive—present in the compositions should not prevent the composition from being readily extrudable from a suitable dispensing apparatus, such as a tube: conventionally such extrudable curable compositions should possess a viscosity of from 3000 to 150,000, preferably from 40,000 to 80,000 mPas, or even from 50,000 to 60,000 mPas.

The above viscosity conditions aside, it is preferred that said electrically non-conductive filler is present in an amount of from 1 to 50% by weight of the total weight of the composition, preferably from 2 to 25% more preferably from 3 to 10%.

These electrically non-conductive filler quantities are preferred because quantity greater than 50% may lead to inadequate adhesion properties, whereas low quantities, mainly below 1%, may lead to inadequate adhesion properties and viscosity issues.

Electrically Conductive Fillers

As noted hereinabove, the two-part (2K) composition of the present invention may comprise electrically conductive fillers, which fillers may be disposed in one or both parts of the composition. The presence, identity and amount of electrically conductive filler in each part is independently determined and, as such, may be the same or different for each part.

Broadly, there is no particular intention to limit the shape of the particles employed as conductive fillers: particles that are acicular, spherical, ellipsoidal, cylindrical, bead-like, cubic or platelet-like may be used alone or in combination. Moreover, it is envisaged that agglomerates of more than one particle type may be used. Equally, there is no particular intention to limit the size of the particles employed as conductive fillers. However, such conductive fillers will conventionally have an average volume particle size, as measured by laser diffraction/scattering methods, of from 1 to 500 µm, for example from 1 to 200 µm.

Exemplary conductive fillers include, but are not limited to: silver; copper; gold; palladium; platinum; nickel; gold or silver-coated nickel; carbon black; carbon fibre; graphite; aluminium; indium tin oxide; silver coated copper; silver coated aluminium; metallic coated glass spheres; metallic

coated filler; metallic coated polymers; silver coated fibre; silver coated spheres; antimony doped tin oxide; conductive nanospheres; nano silver; nano aluminium; nano copper; nano nickel; carbon nanotubes; and, mixtures thereof. The use of particulate silver and/or carbon black as the conductive filler is preferred.

In certain important embodiments, the electrically conductive fillers should be included in the composition in an amount of from 0.05 to 10 wt. %, based on the total weight of the composition. Preferably the composition comprises from 0.1 to 5 wt. %, for example 0.5 to 4 wt. % or from 1 to 3 wt. % of electrically conductive fillers, based on the total weight of the composition.

These electrically conductive particles quantities are preferred because quantity greater than 10% may increase the conductivity, but at the same adversely affect adhesion and debonding properties, whereas low quantities, mainly below 0.05% may lead no physical effect, and a quantity of an electrolyte cannot be decreased.

Tougheners

The presence of tougheners in one or both parts of the present composition can be advantageous. In particular, the first part of the composition of the present invention should preferably contain at least one toughener selected from: epoxy-elastomer adducts; and, toughening rubber in the form of core-shell particles dispersed in the epoxy resin matrix.

Elastomer-containing adducts may be used individually or a combination of two or more particular adducts might be used. Moreover, each adduct may independently be selected from solid adducts or liquid adducts at a temperature of 23° C. Typically, useful adducts will be characterized by a ratio by weight of epoxy to elastomer of from 1:5 to 5:1, for example from 1:3 to 3:1. And an instructive reference regarding suitable epoxy/elastomer adducts is US Patent Publication 2004/0204551. Moreover, exemplary commercial epoxy/elastomer adducts for use herein include but are not limited to: HYPDX RK8-4 commercially available from CVC Chemical; and, B-Tough A3 available from Croda Europe Limited.

The term “core shell rubber” or CSR is being employed in accordance with its standard meaning in the art as denoting a rubber particle core formed by a polymer comprising an elastomeric or rubbery polymer as a main ingredient and a shell layer formed by a polymer which is graft polymerized onto the core. The shell layer partially or entirely covers the surface of the rubber particle core in the graft polymerization process. By weight, the core should constitute at least 50 wt. % of the core-shell rubber particle.

The polymeric material of the core should have a glass transition temperature (T_g) of no greater than 0° C. and preferably a glass transition temperature (T_g) of -20° C. or lower, more preferably -40° C. or lower and even more preferably -60° C. or lower. The polymer of the shell is non-elastomeric, thermoplastic or thermoset polymer having a glass transition temperature (T_g) of greater than room temperature, preferably greater than 30° C. and more preferably greater than 50° C.

Without intention to limit the invention, the core may be comprised of: a diene homopolymer, for example, a homopolymer of butadiene or isoprene; a diene copolymer, for example a copolymer of butadiene or isoprene with one or more ethylenically unsaturated monomers, such as vinyl aromatic monomers, (meth)acrylonitrile or (meth)acrylates; polymers based on (meth)acrylic acid ester monomers, such

as polybutylacrylate; and, polysiloxane elastomers such as polydimethylsiloxane and crosslinked polydimethylsiloxane.

Similarly without intention to limit the present invention, the shell may be comprised of a polymer or copolymer of one or more monomers selected from: (meth)acrylates, such as methyl methacrylate; vinyl aromatic monomers, such as styrene; vinyl cyanides, such as acrylonitrile; unsaturated acids and anhydrides, such as acrylic acid; and, (meth) acrylamides. The polymer or copolymer used in the shell may possess acid groups that are cross-linked ionically through metal carboxylate formation, in particular through forming salts of divalent metal cations. The shell polymer or copolymer may also be covalently cross-linked by monomers having two or more double bonds per molecule.

It is preferred that any included core-shell rubber particles have an average particle size (d_{50}) of from 10 nm to 300 nm, for example from 50 nm to 250 nm: said particle size refers to the diameter or largest dimension of a particle in a distribution of particles and is measured via dynamic light scattering. For completeness, the present application does not preclude the presence of two or more types of core shell rubber (CSR) particles with different particle size distributions in the composition to provide a balance of key properties of the resultant cured product, including shear strength, peel strength and resin fracture toughness.

The core-shell rubber may be selected from commercially available products, examples of which include: Paraloid EXL 2650A, EXL 2655 and EXL2691 A, available from The Dow Chemical Company; Clearstrength® XT100, available from Arkema Inc.; the Kane Ace® MX series available from Kaneka Corporation, and in particular MX 120, MX 125, MX 130, MX 136, MX 551, MX553; and, METABLEN SX-006 available from Mitsubishi Rayon.

In toto tougheners should be included in the composition in an amount of from 0 to 15 wt. %, for example in an amount of from 1 to 10 wt. %, based on the total weight of the composition. In an alternative expression of preference, which is not intended to be mutually exclusive of that mentioned above, tougheners should in toto constitute from 0 to 20 wt. %, preferably from 2 to 15 wt. % of the first part of the composition.

These toughener quantities are preferred because quantity greater than 15% may lead to inadequate adhesion properties, whereas low quantities, mainly below 1%, may lead to inadequate debonding effect and a composition may be too flexible.

Additives and Adjunct Ingredients

Said compositions obtained in the present invention will typically further comprise adjuvants and additives that can impart improved properties to these compositions. For instance, the adjuvants and additives may impart one or more of: improved elastic properties; improved elastic recovery; longer enabled processing time; faster curing time; and, lower residual tack. Included among such adjuvants and additives—which independently of one another may be included in single parts or both parts of a two (2K) part composition—are: plasticizers; stabilizers including UV stabilizers; antioxidants; reactive diluents; drying agents; adhesion promoters; fungicides; flame retardants; rheological adjuvants; color pigments or color pastes; and/or optionally also, to a small extent, non-reactive diluents.

Such adjuvants and additives can be used in such combination and proportions as desired, provided they do not adversely affect the nature and essential properties of the composition. While exceptions may exist in some cases, these adjuvants and additives should not in toto comprise

more than 50 wt. % of the total composition and preferably should not comprise more than 20 wt. % of the composition.

For completeness, it is noted that, in general, adjunct materials and additives which contain reactive groups will be blended into the appropriate part of the two (2K) part composition to ensure the storage stability thereof. Unreactive materials may be formulated into either or both of the two parts.

A “plasticizer” for the purposes of this invention is a substance that decreases the viscosity of the composition and thus facilitates its processability. Herein the plasticizer may constitute up to 10 wt. % or up to 5 wt. %, based on the total weight of the composition, and is preferably selected from the group consisting of: diurethanes; ethers of monofunctional, linear or branched C4-C16 alcohols, such as Cetiol OE (obtainable from Cognis Deutschland GmbH, Dusseldorf); esters of abietic acid, butyric acid, thiobutyric acid, acetic acid, propionic acid esters and citric acid; esters based on nitrocellulose and polyvinyl acetate; fatty acid esters; dicarboxylic acid esters; esters of OH-group-carrying or epoxidized fatty acids; glycolic acid esters; benzoic acid esters; phosphoric acid esters; sulfonic acid esters; trimellitic acid esters; polyether plasticizers, such as end-capped polyethylene or polypropylene glycols; polystyrene; hydrocarbon plasticizers; chlorinated paraffin; and, mixtures thereof. It is noted that, in principle, phthalic acid esters can be used as the plasticizer but these are not preferred due to their toxicological potential.

“Stabilizers” for purposes of this invention are to be understood as antioxidants, UV stabilizers, thermal stabilizers or hydrolysis stabilizers. Herein stabilizers may constitute in toto up to 10 wt. % or up to 5 wt. %, based on the total weight of the composition. Standard commercial examples of stabilizers suitable for use herein include: sterically hindered phenols; thioethers; benzotriazoles; benzophenones; benzoates; cyanoacrylates; acrylates; amines of the hindered amine light stabilizer (HALS) type; phosphorus; sulfur; and, mixtures thereof.

In order to enhance shelf life even further, it is often advisable to further stabilize the compositions of the present invention with respect to moisture penetration through using drying agents. A need also occasionally exists to lower the viscosity of an adhesive or sealant composition according to the present invention for specific applications, by using reactive diluent(s). The total amount of reactive diluents present will typically be from 0 to 15 wt. %, for example from 0 to 5 wt. %, based on the total weight of the composition.

The presence of solvents and non-reactive diluents in the compositions of the present invention is also not precluded where this can usefully moderate the viscosities thereof. For instance, but for illustration only, the compositions may contain one or more of: xylene; 2-methoxyethanol; dimethoxyethanol; 2-ethoxyethanol; 2-propoxyethanol; 2-isopropoxyethanol; 2-butoxyethanol; 2-phenoxyethanol; 2-benzyloxyethanol; benzyl alcohol; ethylene glycol; ethylene glycol dimethyl ether; ethylene glycol diethyl ether; ethylene glycol dibutyl ether; ethylene glycol diphenyl ether; diethylene glycol; diethylene glycol-monomethyl ether; diethylene glycol-monoethyl ether; diethylene glycol-mono-n-butyl ether; diethylene glycol dimethyl ether; diethylene glycol diethyl ether; diethylene glycoldi-n-butyl ether; propylene glycol butyl ether; propylene glycol phenyl ether; dipropylene glycol; dipropylene glycol monomethyl ether; dipropylene glycol dimethyl ether; dipropylene glycoldi-n-butyl ether; N-methylpyrrolidone; diphenylmethane; diisopropyl-naphthalene; petroleum fractions such as

Solvesso® products (available from Exxon); alkylphenols, such as tert-butylphenol, nonylphenol, dodecylphenol and 8,11,14-pentadecatrienylphenol; styrenated phenol; bisphenols; aromatic hydrocarbon resins especially those containing phenol groups, such as ethoxylated or propoxylated phenols; adipates; sebacates; phthalates; benzoates; organic phosphoric or sulfonic acid esters; and sulfonamides.

The above aside, it is preferred that said non-reactive diluents constitute in toto less than 10 wt. %, in particular less than 5 wt. % or less than 2 wt. %, based on the total weight of the composition.

Illustrative Embodiment of the Two-Part (2K) Composition

In an exemplary embodiment of the present invention, the two-part (2K) adhesive composition comprises:

a first part comprising, based on the weight of the composition

from 30 to 70 wt. %, preferably 35 to 60 wt. % of said epoxy resin;

from 2 to 25 wt. %, preferably 5 to 20 wt. % of said electrolyte, wherein said electrolyte is selected from 1-ethyl-3-methylimidazolium methansulfonate, 1-ethyl-3-methylimidazolium methyl sulfate, and mixtures thereof; and,

from 0 to 15 wt. %, preferably from 1.5 to 10 wt. %, of solubilizer, wherein said solubilizer comprises a polyoxy(C₂-C₃)alkylene glycol having a weight average molecular weight of from 200 to 10000 g/mol, for example 200 to 2000 g/mol; and,

a second part comprising:

a curing agent consisting of at least one compound possessing at least two epoxide reactive groups per molecule, wherein said curing agent comprises at least one polyamine having at least two amine hydrogens reactive toward epoxide groups, said polyamine being further characterized by containing primary and/or secondary amine groups and having an equivalent weight per primary or secondary amine group of not more than 150 g/eq;

from 1 to 15 wt. %, based on the weight of the composition, of accelerator, wherein said accelerator is selected from the group consisting of tertiary amines, quaternary ammonium salts, amidines, guanidines and mixtures thereof; and, wherein said composition is further characterized

by comprising from 1 to 50 wt. %, preferably from 2 to 25%, of electrically non-conductive filler.

Methods and Applications

To form the defined two part (2K) curable compositions, the reactive parts are brought together and mixed in such a manner as to induce the hardening thereof. The reactive compounds should be mixed under sufficient shear forces to yield a homogeneous mixture. It is considered that this can be achieved without special conditions or special equipment. That said, suitable mixing devices might include: static mixing devices; magnetic stir bar apparatuses; wire whisk devices; augers; speed mixer; batch mixers; planetary mixers; C.W. Brabender or Banbury® style mixers; and, high shear mixers, such as blade-style blenders and rotary impellers.

For small-scale applications in which volumes of less than 2 litres will generally be used, the preferred packaging for the two part (2K) compositions will be side-by-side double cartridges or coaxial cartridges, in which two tubular chambers—typically of equal volume—are arranged alongside one another or inside one another and are sealed with

pistons: the driving of these pistons allows the parts to be extruded from the cartridge, advantageously through a closely mounted static or dynamic mixer.

For larger volume applications, the two parts of the composition may advantageously be stored in drums or pails: in this case the two parts are extruded via hydraulic presses, in particular by way of follower plates, and are supplied via pipelines to a mixing apparatus which can ensure fine and highly homogeneous mixing of the two parts. In any event, for any package it is important that the parts be disposed with an airtight and moisture-tight seal, so that both parts can be stored for a long time, ideally for 12 months or longer.

Non-limiting examples of two part dispensing apparatuses and methods that may be suitable for the present invention include those described in U.S. Pat. Nos. 6,129,244 and 8,313,006.

The first and second parts of the composition will conventionally be mixed in amounts by weight which achieve the desired ratio of epoxide groups to epoxide-reactive groups. By way of example, the ratio by weight of said first part to said second part may be in the range from 12:1 to 2:1, preferably from 10:1 to 2:1, for example from 8:1 to 3:1 or 7:1 to 5:1.

Where applicable, the two (2K) part curable compositions should broadly be formulated to exhibit an initial viscosity—determined immediately after mixing, for example, up to two minutes after mixing—of less than 200000 mPa·s, for instance less than 100000 mPa·s, at 25° C. Independently of or additional to said viscosity characteristics, the two (2K) part composition should be formulated to be bubble (foam) free upon mixing and subsequent curing.

In accordance with the broadest process aspects of the present invention, the above described compositions are applied to the material layer(s) and then cured in situ. Prior to applying the compositions, it is often advisable to pre-treat the relevant surfaces to remove foreign matter therefrom: this step can, if applicable, facilitate the subsequent adhesion of the compositions thereto. Such treatments are known in the art and can be performed in a single or multi-stage manner constituted by, for instance, the use of one or more of: an etching treatment with an acid suitable for the substrate and optionally an oxidizing agent; sonication; plasma treatment, including chemical plasma treatment, corona treatment, atmospheric plasma treatment and flame plasma treatment; immersion in a waterborne alkaline degreasing bath; treatment with a waterborne cleaning emulsion; treatment with a cleaning solvent, such as carbon tetrachloride or trichloroethylene; and, water rinsing, preferably with deionized or demineralized water. In those instances where a waterborne alkaline degreasing bath is used, any of the degreasing agent remaining on the surface should desirably be removed by rinsing the substrate surface with deionized or demineralized water.

The compositions are then applied to the preferably pre-treated surfaces of the substrate by conventional application methods such as: brushing; roll coating using, for example, a 4-application roll equipment where the composition is solvent-free or a 2-application roll equipment for solvent-containing compositions; doctor-blade application; printing methods; and, spraying methods, including but not limited to air-atomized spray, air-assisted spray, airless spray and high-volume low-pressure spray.

As noted above, the present invention provides a bonded structure comprising: a first material layer having an electrically conductive surface; and, a second material layer having an electrically conductive surface, wherein the cured

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and debondable two-part (2K) adhesive composition as defined hereinabove and in the appended claims is disposed between said first and second material layers. To produce such a structure, the adhesive composition may be applied to at least one internal surface of the first and/or second material layer and the two layers then subsequently contacted, optionally under the application of pressure, such that the electrically debondable hot melt adhesive composition is interposed between the two layers.

It is recommended that the compositions be applied to a surface at a wet film thickness of from 10 to 500 μm . The application of thinner layers within this range is more economical and provides for a reduced likelihood of deleterious thick cured regions. However, great control must be exercised in applying thinner coatings or layers so as to avoid the formation of discontinuous cured films.

The curing of the applied compositions of the invention typically occurs at temperatures in the range of from 40° C. to 200° C., preferably from 50° C. to 175° C., and in particular from 75° C. to 175° C. The temperature that is suitable depends on the specific compounds present and the desired curing rate and can be determined in the individual case by the skilled artisan, using simple preliminary tests if necessary. Of course, curing at lower temperatures within the aforementioned ranges is advantageous as it obviates the requirement to substantially heat or cool the mixture from the usually prevailing ambient temperature. Where applicable, however, the temperature of the mixture formed from the respective parts of a two (2K) part composition may be raised above the mixing temperature and/or the application temperature using conventional means including microwave induction.

The present invention will be described with reference to the appended drawings in which:

FIG. 1a illustrates a bonded structure in accordance with a first embodiment of the present invention.

FIG. 1b illustrates a bonded structure in accordance with a second embodiment of the present invention.

FIG. 2a illustrates the initial debonding of the structure of the first embodiment upon passage of a current across that structure.

FIG. 2b illustrates the initial debonding of the structure of the second embodiment upon passage of a current across that structure.

FIGS. 3a and 3b illustrate the results of lap shear strength testing of aluminium substrates bonded with a cured adhesive composition in accordance with an embodiment of the present invention.

FIG. 4 illustrates adhesion properties and debonding effect on aluminium and steel in terms of strength (measured in MPa) over storage time (measured in days).

FIG. 5 illustrates a bar graph of the strength (measured in MPa) both initially and after application of voltage in Samples with and without fumed silica, demonstrating the improved strength and debonding effect after application of voltage.

As shown in FIG. 1a appended hereto, a bonded structure is provided in which a layer of cured adhesive (10) is disposed between two conductive substrates (11). A layer of non-conductive material (12) may be disposed on the conductive substrates (11) to form the more complex bonded structure as depicted in FIG. 1b. Each layer of conductive substrate (11) is in electrical contact with an electrical power source (13) which may be a battery or an AC-driven source of direct current (DC). The positive and negative terminals of that power source (13) are shown in one fixed position but

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the skilled artisan will of course recognize that the polarity of the system can be reversed.

The two conductive substrates (11) are shown in the form of a layer which may be constituted by inter alia: a metallic film; a metallic mesh or grid; deposited metal particles; a resinous material which is rendered conductive by virtue of conductive elements disposed therein; or, a conducting oxide layer. As exemplary conductive elements there may be mentioned silver filaments, single-walled carbon nanotubes and multi-walled carbon nanotubes. As exemplary conducting oxides there may be mentioned: doped indium oxides, such as indium tin oxide (ITO); doped zinc oxide; antimony tin oxide; cadmium stannate; and, zinc stannate. The selection of the conductive material aside, the skilled artisan will recognize that the efficacy of the debonding operation may be diminished where the conductive substrates (11) are in the form of a grid or mesh which offers limited contact with the layer of cured adhesive (10).

As depicted in FIGS. 2a and 2b, the debonding occurs at the positive interface, that interface between the adhesive composition (10) and the electrically conductive surface (11) that is in electrical contact with the positive electrode. By reversing current direction prior to separation of the substrates, the adhesive bond may be weakened at both substrate interfaces.

It is however noted that the composition of the adhesive layer (10) may be moderated so that debonding occurs at either the positive or negative interface or simultaneously from both. For some embodiments, a voltage applied across both surfaces so as to form an anodic interface and a cathodic interface will cause debonding to occur simultaneously at both the anodic and cathodic adhesive/substrate interfaces. In an alternative embodiment, reversed polarity may be used to simultaneously disbond both substrate/adhesive interfaces if the composition does not respond at both interfaces to direct current. The current can be applied with any suitable waveform, provided that sufficient total time at each polarity is allowed for debonding to occur. Sinusoidal, rectangular and triangular waveforms might be appropriate in this regard and may be applied from a controlled voltage or a controlled current source.

Without intention to limit the present invention, it is considered that the debonding operation may be performed effectively where at least one and preferably both of the following conditions are instigated: a) an applied voltage of from 0.5 to 200 V; and, b) the voltage being applied for a duration of from 1 second to 120 minutes, for example from 1 second to 60 minutes or 1 second to 30 minutes. Where the release of the conductive substrate from the cured adhesive is to be facilitated by the application of a force—exerted via a weight or a spring, for instance—the potential might only need to be applied for the order of seconds.

The following examples are illustrative of the present invention and are not intended to limit the scope of the invention in any way.

EXAMPLES

The following materials were employed in the Examples: Cab-O-Sil 720: Fumed silica which has been surface treated with polydimethylsiloxane (PDMS) available from Cabot Corporation. Casiflux G20: Particulate wollastonite, available from Sibelco. KBM-4803: 8-glycidooxyoctyl trimethoxysilane, available from Shinetsu.

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EMIM-MS 1-Ethyl-3-methylimidazolium meth-
ansulfonate, available from TCI America Inc.

B-Tough A3 Epoxy-silicone elastomer adduct, available
from Croda Europe Limited.

DER 330EL: Low viscosity bisphenol A epoxy resin,
available from Olin.

DER 337: Intermediate epoxy equivalent weight semi-
solid resin based on Bisphenol-A epoxy, available from
Olin.

DER 337-X80: Intermediate epoxy equivalent weight
semi-solid resin based on Bisphenol-A epoxy in xylene,
available from Olin.

Jeffamine® D-230: Ether group-containing aliphatic pri-
mary polyamine, available from Huntsman.

PEG400: Polyethylene glycol, available from Sigma
Aldrich.

2-piperazin-1-ylethanamine Tertiary amine, available
from Acros Organics.

Example 1

Parts (A) and (B) of the two-part (2K) compositions were
independently prepared in accordance with Table 1 herein
below:

TABLE 1

Ingredient		Wt. % (based on weight of composition)
Part A		
1	D.E.R. 337	52.81
2	B-tough A3	4.18
3	Casiflux G20	4.18
4	EMIM MS	18
5	PEG 400	4.57
6	ShinEtsu KBM-4803	0.76
7	Cab-o-Sil TS 720	1.52
Part B		
1	2-piperazin-1-ylethanamine	6.99
2	Jeffamine D230	6.99

The individual parts (A, B) were loaded into separate
compartments of a 50 g cartridge and sealed at both ends.
The cartridge was then loaded into a cartridge-gun and a
mixing tip was installed on the front end. By application of
constant pressure on the trigger, the two parts were pushed
into the mixing tip to ensure sufficient mixing before appli-
cation to the stated substrate.

The substrates were aluminium (AA6016) and stainless
steel (1.4301) each with a thickness of 1.25 mm for alu-
minium and 1.5 mm for steel and 1 mm for copper. Sub-
strates were cut into 2.5 cm×10 cm (1"×4") in size for tensile
testing. Tensile lap shear (TLS) test was performed accord-
ing to test method described on page 5.

The applied two-part (2K) adhesive compositions were
cured in the overlapping region by the application of a
temperature of 65° C. for 120 minutes. Subsequently, the
samples were stored at 25° C. with 20% humidity in a
climate chamber.

For each substrate, tensile lap shear strength was inves-
tigated after said 24-hour storage period both prior and
subsequent to the application of a constant potential of 50V
across the adhesive layer for a duration of 20 minutes. The
results are documented in Table 2 herein below.

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TABLE 2

Substrate	Initial Bond Strength (MPa)	Bond Strength after 50 V, 20 minutes (MPa)
Aluminium	15.16 (±1.79)	5.9 (±1.9)
Stainless Steel	19.11 (±0.88)	5.14 (±1.42)
Copper	15.21 (±0.48)	1.88 (±0.28)

Example 2

For the adhesively bonded aluminium substrate
(AA6016), lap shear strength (MPa) was investigated under
two conditions: a) over a period of 120 minutes applying a
constant potential (75 V) across the overlapping bonded
area; and, b) applying different potentials across the over-
lapping bonded region for a fixed period of time (30 min-
utes) at each applied potential. The results of these investi-
gations are given in FIGS. 3a and 3b appended hereto.

Example 3

Stability test was conducted for the composition of
example 1. For this test, normal lap shear samples were
prepared and cured at 65° C. for 120 min. Aluminium and
steel substrates were used. Subsequently, the samples were
stored at 25° C. with 20% humidity in a climate chamber.

The las shear was measured after one day, seven days, 14
days, 21 days and 30 days. The results are documented in
Table 3 herein below.

TABLE 3

	Initial Bond Strength (MPa)	Bond Strength after 75 V, 1 hour (MPa)
Aluminium		
1 day	15.155 (+/-) 1.79	5.9 (+/-) 1.89
7 days	16.8 (+/-) 0.68	4.97 (+/-) 2.12
14 days	17.4 (+/-) 1.19	3.04 (+/-) 0.17
21 days	12.11 (+/-) 0.79	4.54 (+/-) 0.82
30 days	14.57 (+/-) 2.21	4.32 (+/-) 0.42

The stability results are illustrated in FIG. 4. FIG. 4
illustrates adhesion properties and debonding effect on alu-
minium and steel. The test results show that the composition
according to the present invention has good initial adhesion
properties and does not lose them over the time. In addition,
the composition according to the present invention has good
initial debonding effect and maintains it over the time.

Example 4

Parts (A) and (B) of the two-part (2K) compositions were
independently prepared in accordance with Table 4 herein
below:

TABLE 4

Ingredient		Wt. % (based on weight of composition)
Part A		
1	D.E.R. 330-EL	52.81
2	B-tough A3	4.18
3	Casiflux G20	4.18
4	EMIM MS	18.00
5	PEG 400	4.57
6	ShinEtsu KBM-4803	0.76
7	Cab-o-Sil TS 720	1.52

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TABLE 4-continued

Ingredient		Wt. % (based on weight of composition)
Part B		
1	1-(2-aminoethyl)piperazin	6.99
2	Jeffamine D230	6.99

The composition and samples were prepared as in described in example 1. The results are documented in Table 5 herein below.

TABLE 5

Substrate	Initial Bond Strength (MPa)	Bond Strength after 50 V, 20 minutes (MPa)
Aluminium	18.20 (± 0.925)	10.02 (± 0.8)
Stainless Steel	27.38 (± 0.81)	16.31 (± 4.99)

In view of the foregoing description and examples, it will be apparent to those skilled in the art that equivalent modifications thereof can be made without departing from the scope of the claims.

Example 5

Parts (A) and (B) of the two-part (2K) compositions were independently prepared in accordance with Table 6 herein below:

TABLE 6

Ingredient		Wt. % (based on weight of composition)
Part A		
1	D.E.R. 337-X80	55.83
2	B-tough A3	4.18
3	Casiflux G20	4.18
4	EMIM MS	18.00
5	PEG 400	4.57
6	ShinEtsu KBM-4803	0.76
7	Cab-o-Sil TS 720	1.52
Part B		
1	1-(2-aminoethyl)piperazine	5.48
2	Jeffamine D230	5.48

The composition and samples were prepared as in described in example 1. The results are documented in Table 7 herein below.

TABLE 7

Substrate	Initial Bond Strength (MPa)	Bond Strength after 50 V, 20 minutes (MPa)
Aluminium	6.43 (± 0.75)	0.95 (± 0.23)

Example 6

Parts (A) and (B) of the two-part (2K) compositions were independently prepared in accordance with Table 8 herein below:

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TABLE 8

Ingredient		Wt. % (based on weight of composition)
Part A		
1	D.E.R. 337	61.63
2	B-tough A3	4.72
3	Casiflux G20	4.72
4	EMIM MS	8.77
5	PEG 400	5.15
6	ShinEtsu KBM-4803	0.86
7	Cab-o-Sil TS 720	1.72
8	GPX 801 (Cabot)	0.09
Part B		

1	1-(2-aminoethyl)piperazine	6.17
2	Jeffamine D230	6.17

The composition and samples were prepared as in described in example 1. The results are documented in Table 9 herein below.

TABLE 9

Substrate	Initial Bond Strength (MPa)	Bond Strength after 50 V, 20 minutes (MPa)
Aluminium	17.69 (± 1.64)	3.21 (± 0.91)

In view of the foregoing description and examples, it will be apparent to those skilled in the art that equivalent modifications thereof can be made without departing from the scope of the claims.

Example 7

Two two-part (2K) compositions were independently prepared in accordance with Table 10 herein below with the difference that the composition 1a contains a non-electrically conductive filler (fumed silica) whereas composition 1b does not contain a non-electrically conductive filler.

TABLE 10

Ingredient	Wt. % (based on weight of composition)	Wt. % (based on weight of composition)
	Composition 1a	Composition 1b
Part A		
1	D.E.R. 337	61.63
2	B-tough A3	4.72
3	Casiflux G20	4.72
4	EMIM MS	20.48
5	PEG 400	5.15
6	ShinEtsu KBM-4803	0.86
7	Cab-o-Sil TS 720	1.72
Part B		
1	1-(2-aminoethyl)piperazine	6.17
2	Jeffamine D230	6.17

The composition and samples were prepared as in described in example 1. The results are documented in Table 11 herein below.

TABLE 11

Substrate	Initial Bond Strength (MPa)	Bond Strength after 75 V, 20 minutes (MPa)
Aluminium	1a	16.17
Aluminium	1b	12.89
		5.91
		11.07

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The results indicate that the use of non-conductive filler, and especially fumed silica, improves the initial strength and debonding effect. The results are also illustrated in FIG. 5.

In view of the foregoing description and examples, it will be apparent to those skilled in the art that equivalent modifications thereof can be made without departing from the scope of the claims.

What is claimed is:

1. A curable and debondable two-part adhesive composition comprising, based on the total weight of the composition:

a first part comprising:

- a a) at least one epoxy resin present in an amount of from 50 to 70 wt %;
- a b) at least one electrolyte present in an amount of from 15 to 25 wt %;
- c) at least one solubilizer present in an amount of from 0.5 to about 15 wt %; and
- d) at least one electrically non-conductive filler comprising fumed silica a present in an amount of from 0.5 to 5 wt %;

a second part comprising:

- a) at least one curing agent comprising at least one compound having at least two epoxide reactive groups per molecule, the at least one curing agent being present in an amount of from 5 to 10 wt %; and
- b) at least one accelerator present in an amount of from 5 to 10 wt %; and

a toughener.

2. The curable and debondable two-part adhesive composition of claim 1, wherein the at least one epoxy resin is selected from the group consisting of bis-phenol A epoxy resin, bis-phenol F epoxy resin, a mixture of bis-phenol A epoxy resin and bis-phenol F epoxy resin, cycloaliphatic epoxy resin, and mixtures thereof.

3. The curable and debondable two-part adhesive composition of claim 1, wherein the at least one electrolyte is selected from the group consisting of 1-ethyl-3-methylimidazolium methanesulfonate, 1-ethyl-3-methylimidazolium methyl sulfate, 1-hexyl-3-methylimidazolium 2-(2-fluoroanilino)-pyridinate, 1-hexyl-3-methylimidazolium imide, 1-butyl-1-methyl-pyrrolidinium 2-(2-fluoroanilino)-pyridinate, 1-butyl-1-methyl-pyrrolidinium imide, trihexyl(tetradecyl) phosphonium 2-(2-fluoroanilino)-pyridinate, cyclohexyltrimethylammonium bis(trifluoromethylsulfonyl) imide, di(2-hydroxyethyl) ammonium trifluoroacetate, N,N-dimethyl (2-hydroxyethyl) ammonium octanoate, methyltriocetylammmonium bis(trifluoromethylsulfonyl) imide, N-ethyl-N—N—N—N-tetramethylguanidinium trifluoromethanesulfonate, guanidinium trifluoromethanesulfonate, 1-butyl-4-methylpyridinium bromide, 1-butyl-3-methylpyridinium tetrafluoroborate, 1-butyl-3-hydroxymethylpyridinium ethylsulfate, 1-butyl-1-methylpyrrolidinium bis(trifluoromethylsulfonyl) imide, 1-butyl-methylpyrrolidinium tris(pentafluoroethyl) trifluorophosphate, 3-methyl imidazolium ethylsulfate, 1-ethyl-3-methylimidazolium chloride, 1-ethyl-3-ethyl-methylimidazolium bromide, 1-butyl-3-methylimidazolium chloride, 1-hexyl-3-methylimidazolium chloride, 1-octyl-3-methylimidazolium chloride, 1-methyl-3-octylimidazolium chloride, 1-propyl-3-methylimidazolium iodide, 1-butyl-3-methylimidazolium tetrafluoroborate, 1-butyl-3-methylimidazolium trifluoromethanesulfonate, 1-butyl-3-methylimidazolium hexafluorophosphate, 1-butyl-2,3-dimethylimidazolium tetrafluoroborate, 1-butyl-2,3-dimethylimidazolium hexafluorophosphate, 1-butylimidazol, 1-methylimidazolium tetrafluoroborate,

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tetrabutylphosphonium tris(pentafluoroethyl) trifluorophosphate, trihexyl(tetradecyl) phosphonium tetrafluoroborate, and mixtures thereof.

4. The curable and debondable two-part adhesive composition of claim 1, wherein the at least one solubilizer is selected from the group consisting of polyphosphazenes, polymethylenesulfides, polyoxyalkylene glycols, polyethylene imines, silicone surfactants, fluorinated silicone surfactants, co-polymers of functionalized polyalkylsiloxanes, epoxy resins, polyhydric alcohols, and sugars.

5. The curable and debondable two-part adhesive composition of claim 1 further comprising electrically conductive particles selected from the group consisting of carbon black, silver, and mixtures thereof.

6. The curable and debondable two-part adhesive composition of claim 1, further comprising electrically conductive particles in an amount of from 0.05 to 10% wt %, based on the total weight of the composition.

7. The curable and debondable two-part adhesive composition of claim 1, wherein the curing agent comprises at least one polyamine having at least two amine hydrogens reactive toward epoxide groups, primary and/or secondary amine groups, and an equivalent weight per primary or secondary amine group of not more than 150 g/eq.

8. The curable and debondable two-part adhesive composition of to claim 1, wherein the accelerator is selected from the group consisting of tertiary amines, quaternary ammonium salts, amidines, guanidines, and mixtures thereof.

9. The curable and debondable two-part adhesive composition of claim 1, wherein the accelerator is selected from the group consisting of imidazole, methylimidazole, benzylidimethylamine, 1,8-diazabicyclo[5.4.0]undec-7-ene, 1,4-diazabicyclo(2,2,2) octane, and mixtures thereof.

10. The curable and debondable two-part adhesive composition of claim 1, wherein the at least one electrolyte comprises a sulfonate.

11. The curable and debondable two-part adhesive composition of claim 1, wherein the at least one electrolyte comprises a sulfite.

12. The curable and debondable two-part adhesive composition of claim 1, wherein the at least one solubilizer comprises poly alkylene glycol.

13. The curable and debondable two-part adhesive composition of claim 1, wherein the composition has an initial bond strength of at least 16.17 MPa.

14. The curable and debondable two-part adhesive composition of claim 1, further comprising at least one additive.

15. The curable and debondable two-part adhesive composition of claim 1, further comprising at least one adjuvant.

16. The curable and debondable two-part adhesive composition of claim 1, further comprising polyethylene glycol.

17. The curable and debondable two-part adhesive composition of claim 1, further comprising an elastomer epoxy resin.

18. A bonded structure comprising

- a first material layer having an electrically conductive surface;
- a second material layer having an electrically conductive surface; and
- a cured debondable two-part adhesive composition of claim 1 disposed between the first and second material layers.

19. A method of debonding the bonded structure of claim 18, the method comprising:

applying a voltage across the first and second material layers to form an anodic interface and a cathodic interface; and

debonding the first and second material layers, wherein the applied voltage is from 0.5 to 200 V. 5

20. A curable and debondable two-part adhesive composition comprising, based on the total weight of the composition:

a first part comprising:

a) bisphenol A epoxy present in an amount of from 50 10 to 70 wt %;

b) 1-ethyl-3-methylimidazolium methansulfonate present in an amount of from 15 to 25 wt %;

c) 8-glycidooxyoctyl trimethoxysilane present in an amount of from 0.5 to 15 wt %; and 15

d) fumed silica present in an amount of from 0.5 to 5 wt %; and

a second part comprising:

a) 2-piperazin-1-ylethanamine present in an amount of from 5 to 10 wt %; and 20

b) an ether group-containing aliphatic primary polyamine present in an amount of from 5 to 0.10 wt %; and

a toughener.

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